SpectraMaster

محوّل ملفات أطياف أشعة جاما

User Manual

Gamma-spectrum file conversion, viewing and analysis

Version 1.0.0 · Windows

Designed & developed by

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1. Introduction and installation

1.1 What SpectraMaster does

SpectraMaster is a gamma-spectrum **file converter, viewer and analysis tool** for the laboratory. It works in the style of ORTEC *DataMaster* (A49-B32): the same "open – choose output format – save" workflow, the same output-file naming rule and the same command line (`-i input -j output` with the DataMaster format numbers). On top of that it offers:

- reading of **59 spectrum formats** (Aptec, ORTEC, Canberra/Genie, IAEA, N42, Excel, many legacy lab formats) and writing of 35 of them, one file at a time or whole folder trees;
- a fast interactive spectrum plot (zoom, history, overview strip, ROIs, overlays, image export);
- automatic **peak search and nuclide identification** with a library built from IAEA ENSDF data;
- energy recalibration, spectrum sum / background subtraction / rebinning (command line);
- **efficiency calibration** and **activity** calculation (Bq, Bq/kg, uncertainty budget, MDA after Currie or ISO 11929);
- printed and PDF **reports** with templates, also for many files at once;
- **archive search** across tens of thousands of old spectra (text, nuclides, date, detector);
- **detector quality control** with control charts and Westgard rules (ISO/IEC 17025, 7.7).

The interface is fully bilingual (Arabic and English) and remembers your choices between sessions.

1.2 Installation

SpectraMaster consists of two programs. Python or any other runtime is **not** needed.

SpectraMaster.exe	The graphical program (windows, plots, analysis, reports). Started with the DataMaster arguments `-i in -j out` it converts silently without opening a window.
smconvert.exe	The command-line converter and tool box for batch files, scripts and the Windows task scheduler (chapter 4 and appendix B).

Installer (recommended)

1. Run `SpectraMaster_Setup_1.0.0.exe` (administrator rights are needed) and choose the language of the setup wizard (English or Arabic).
2. Accept the license, keep the folder `C:\Program Files\SpectraMaster` and choose the components: main program, command-line tool `smconvert` (optionally added to the system `PATH`), user manuals and sample spectra.
3. Choose the tasks: desktop icon and the file associations (`.s0 .chn .cnf .spe .spc .iec .n42`, each can be switched off). Associated files open in SpectraMaster with a double-click; all of them also get **Open with › SpectraMaster**.

The Start menu group **SpectraMaster** contains the program, both user manuals, the release notes, the sample spectra folder and the uninstaller. To pin SpectraMaster to the taskbar, right-click its Start-menu entry (or the running program's taskbar button) and choose **Pin to taskbar**.

Uninstall with **Settings** › **Apps** or the Start-menu entry. The uninstaller asks whether your own data (archive index, efficiency curves, QC tracks in `%APPDATA%\SpectraMaster` and the settings) should be deleted too; the default answer keeps them. It removes only SpectraMaster's own file associations and `PATH` entry.

Portable copy

The two programs are also available as single packed executables: copy them into any folder (even a USB stick) and start `SpectraMaster.exe`; nothing has to be installed. The first start after a reboot can take a few seconds while Windows unpacks the program.

> **Note.** The programs are not code-signed, so Windows SmartScreen may ask for confirmation the first time (**More info** › **Run anyway**, after checking with your IT department).

1.3 First start, splash screen and language

At start-up a splash screen shows the product logo, the designer and the version for about 1.5 seconds (click it to close it at once). It can be switched off with **Help** › **Show Splash Screen at Startup**.



Figure 1 — Splash screen.

The program starts in **English** the first time. Switch the language with **View** › **Language** › **English / العربية**; the whole interface changes at once (including the right-to-left layout) and the choice is remembered. The spectrum plot always runs left to right (low energy on the left) in both languages.

2. The main window

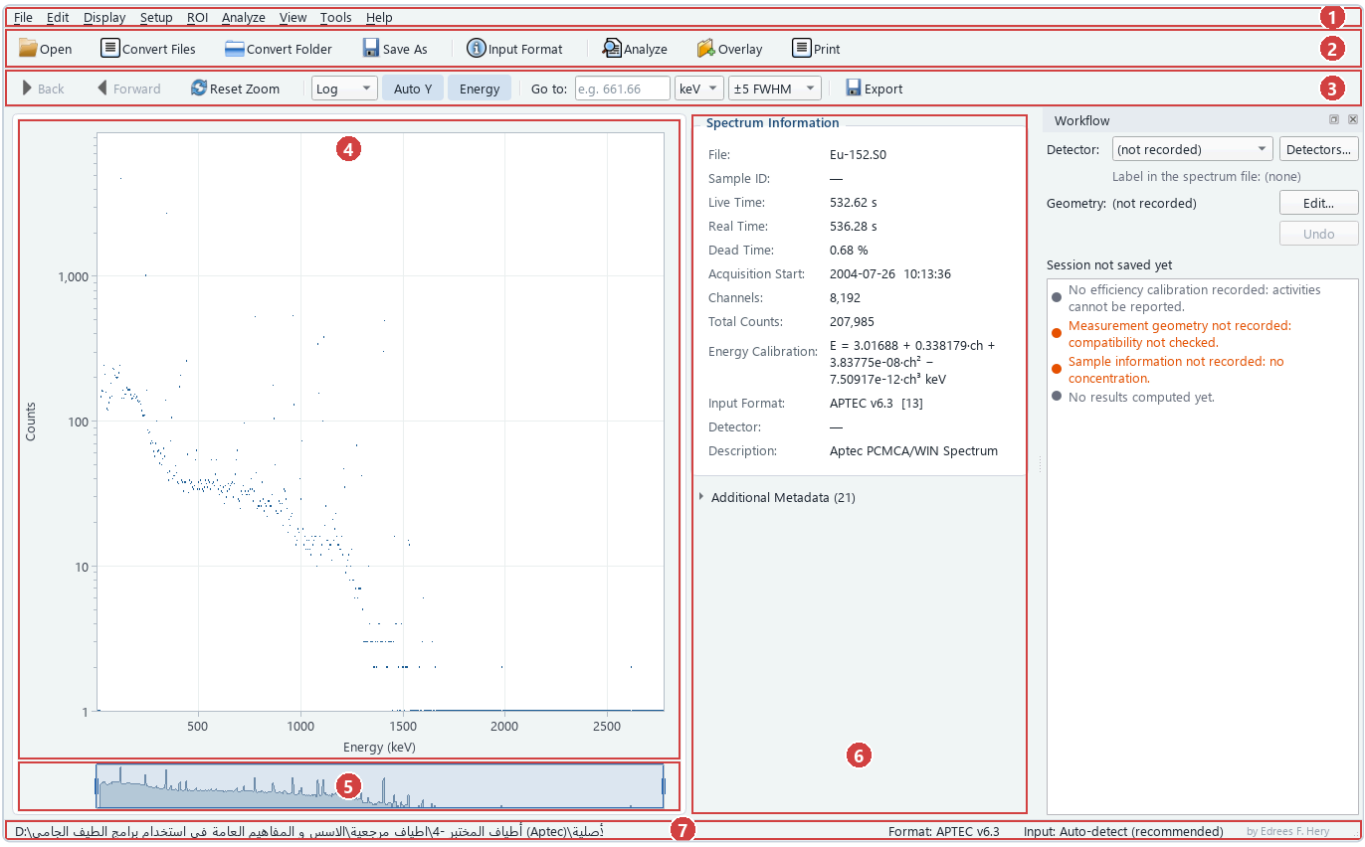


Figure 2 — The main window with a calibration spectrum (Eu-152, Aptec S0) on a logarithmic scale.

1. **Menu bar** — File, Edit, Display, Setup, ROI, Analyze, View, Tools, Help (Aptec's order, chapter 7).
2. **Main toolbar** — Open, Convert Files, Convert Folder, Save As, Input Format, Analyze, Overlay, Print.
3. **Navigation toolbar** — zoom Back / Forward, Reset Zoom, Y scale (Linear / Log / √), Auto Y, Energy axis, *Go to* energy or channel with window width, Export image.
4. **Spectrum plot** — counts versus energy (keV) or channel; mouse read-out in the status bar.
5. **Overview strip** — the whole spectrum with the visible window; drag it to pan.
6. **Spectrum information** — file, sample ID, live/real/dead time, start time, channels, total counts, energy calibration, input format, detector, description and all additional metadata. For files that contain several spectra a selector "Spectrum i of n" appears here.
7. **Status bar** — file path or messages, cursor position (channel, keV, counts), current format, input mode (auto-detect or a fixed format).

2.1 Menus

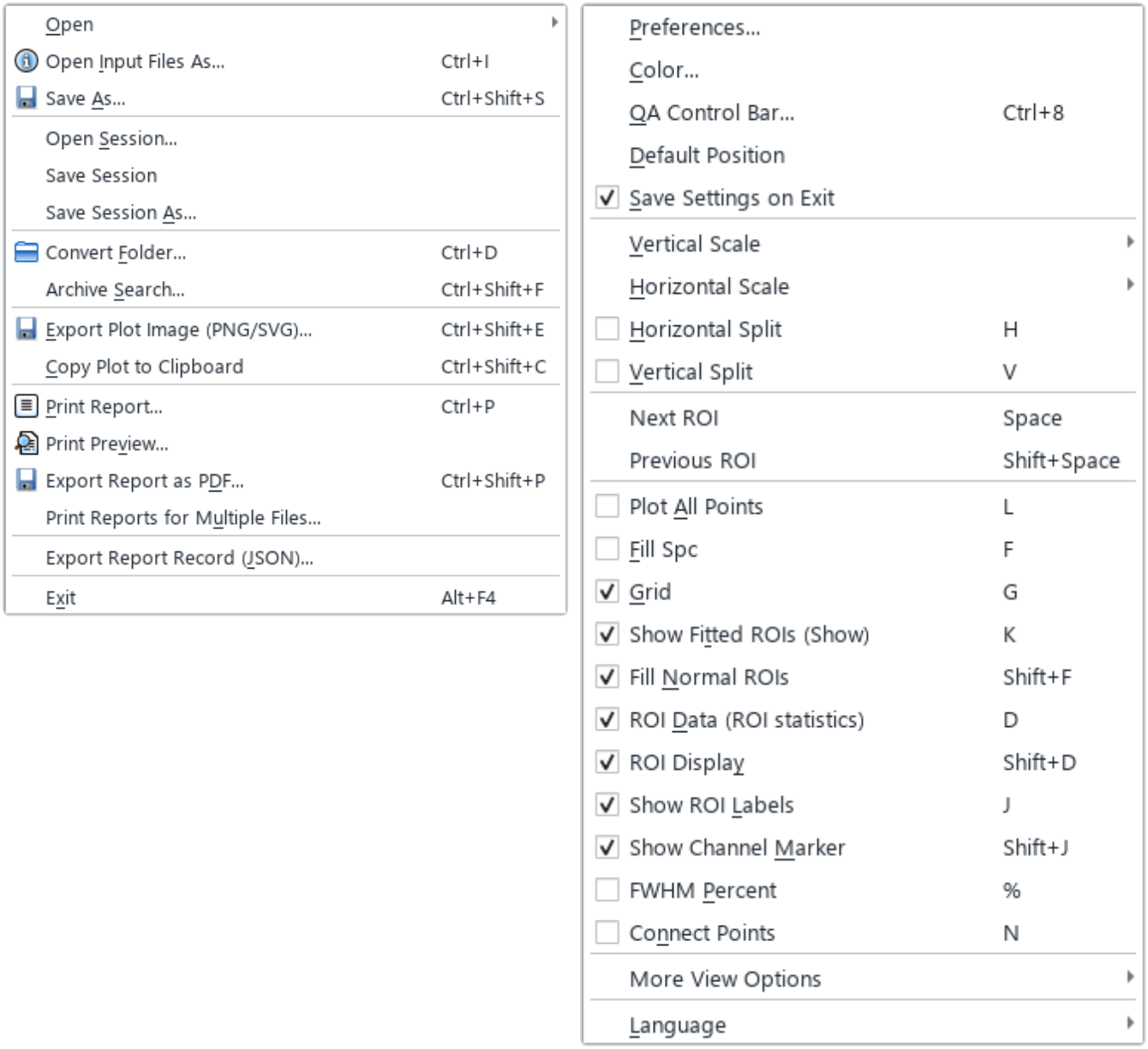


Figure 3 — The File and View menus.

File	Open (spectrum / multiple files), Open Input Files As, Save As, Convert Folder, Archive Search, Export / Copy plot image, Print, Print Preview, Export Report as PDF, Print Reports for Multiple Files, Exit.
Edit	Selection and Copy to ClipBrd, Sample Description (F5), Quantity Description (F6), Spectrum Data, Certificate, Library, Decay Correction Calculation, Load Header Information, Reset Header (section 7.7).
Display	Normal spectrum; energy, resolution and efficiency calibration curves.
Setup	Quick Energy Calibration (F2), full energy / resolution / efficiency calibration (Shift + F2).
ROI	Properties, Paint (F3), Edit (F4), Delete Indexed (Del), Delete All, Sort, Flag Definitions (section 7.3).
Analyze	Setup ›, Peak Search (P), Multifit (M), Identify (I), Background Subtract (B), Quantify (Q); then SpectraMaster's tools (auto-analyze, labels, efficiency and activity, calibration, fit inspection, batch analysis); then the spectrum operations (section 7.5).
View	Preferences, Color, QA bar, split display, spectrum and ROI drawing options, scales, marker and window movement, overlays, panels, language (section 7.6).
Tools	Detector QC.
Help	User Manual (Arabic / English, F1), Mouse & Keyboard Shortcuts, GammaVision / MAESTRO Integration, splash-screen option, About SpectraMaster (version, designer and the full list of formats).

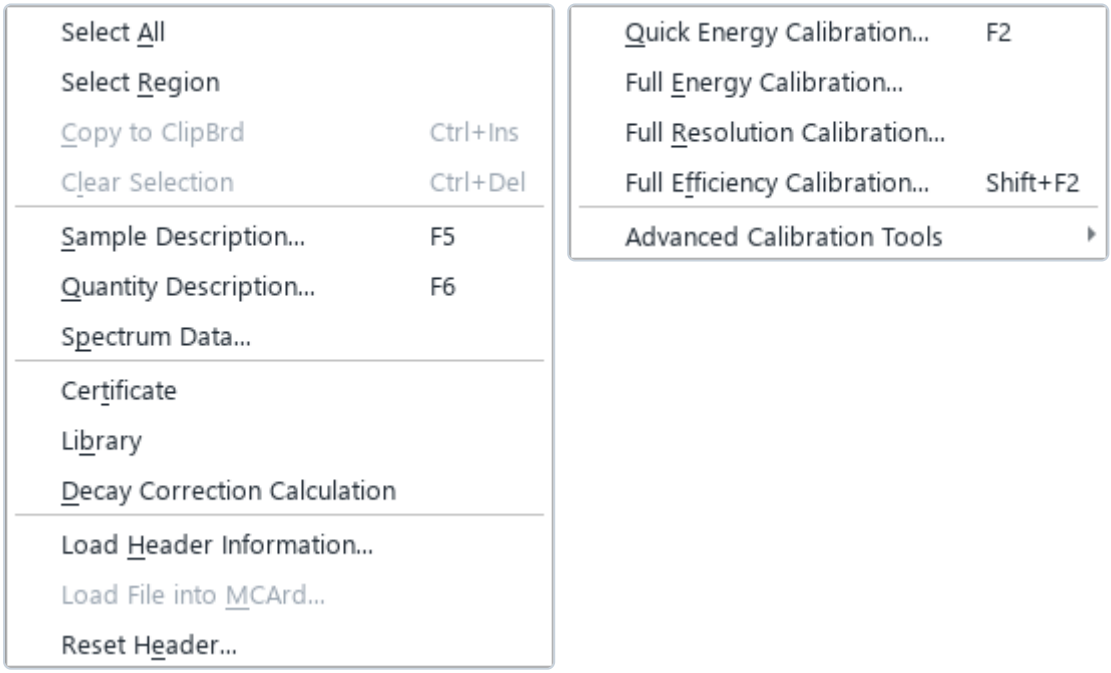


Figure 4 — The Edit and Setup menus.

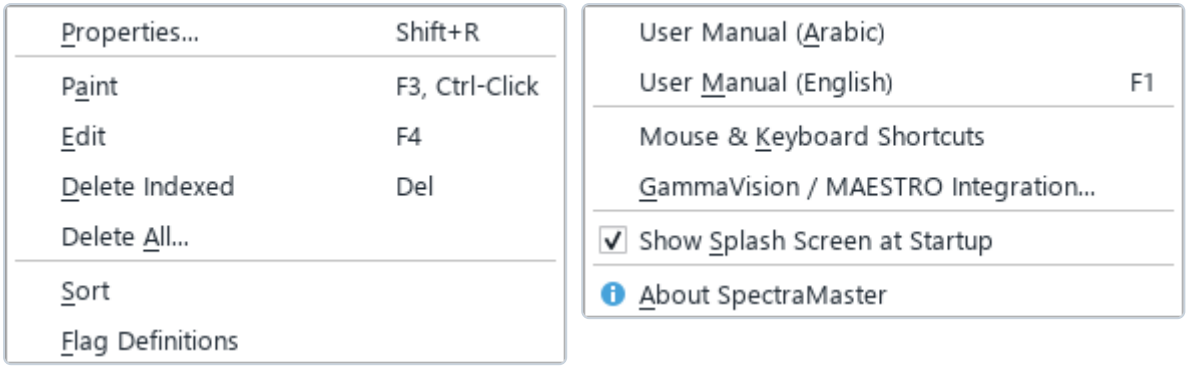


Figure 5 — The ROI and Help menus. The Analyze menu is in section 7.4.

A right click on the plot opens a context menu with the zoom history, Y scale, ROI deletion, analysis, labels, overlay and image export.

3. Opening spectra

3.1 Open a spectrum

1. Choose **File** › **Open** › **Spectrum…** (Ctrl + O) or the **Open** button.
2. Select the file. With **Show description** ticked, the panel on the right shows the detected format, sample ID, description, channels, live/real time, start time and a thumbnail of the spectrum *before* you open it — useful when browsing archives with cryptic file names.
3. Click **Open**. The spectrum is drawn and its details appear in the information panel.

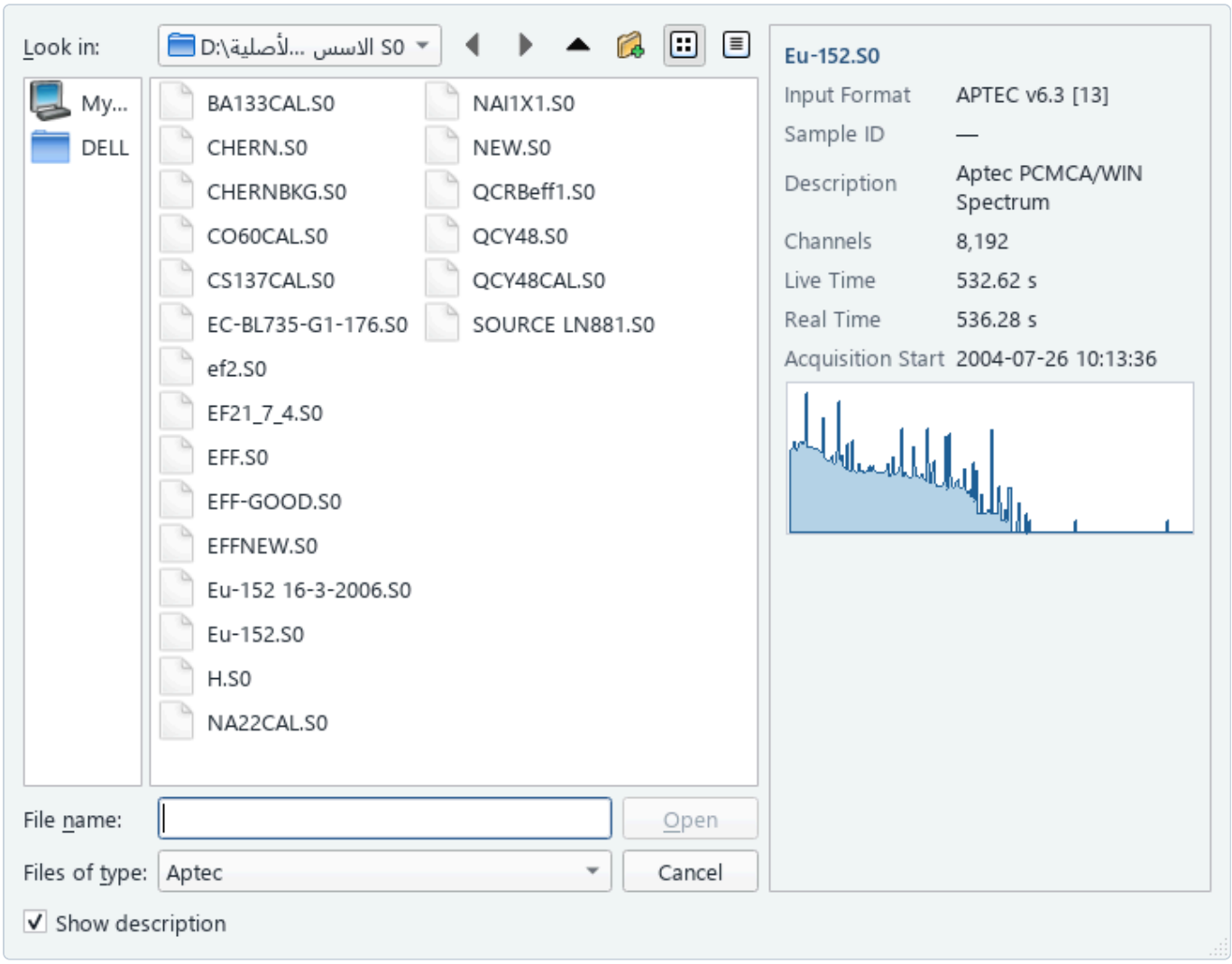


Figure 6 — Open dialog with the description preview.

The file filter lists all readable extensions; choose *All files (*)* for files with unusual extensions — the content is recognised, not the name.

3.2 Input format: auto-detect or fixed

By default the input format is **detected automatically** from the file content (binary signatures and header keywords), then from the extension. **File** › **Open Input Files As…** (Ctrl + I) lets you force one format, exactly

like DataMaster. The choice applies to Open, batch conversion and overlays and is shown in the status bar (*Input: …*).

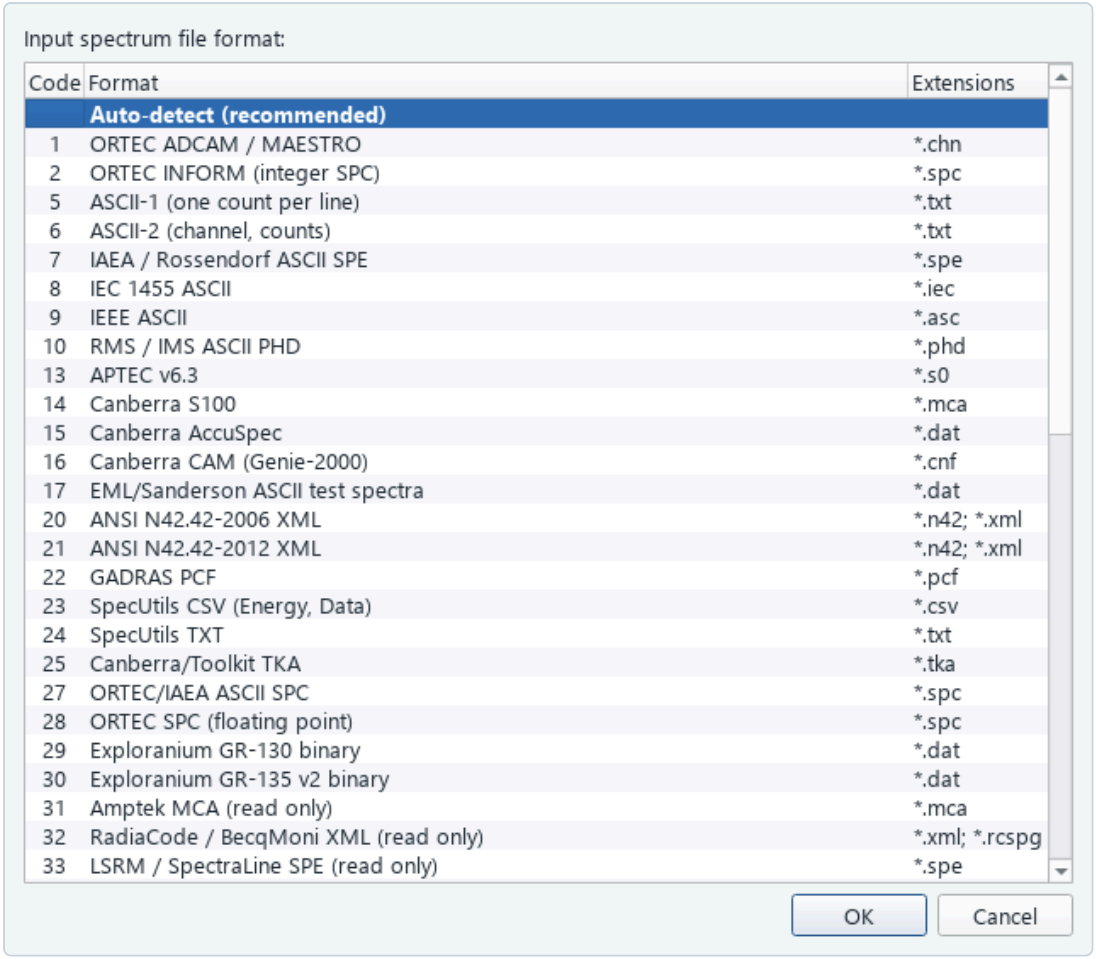


Figure 7 — Choosing the input format (keep *Auto-detect* unless a file is not recognised).

> **Tip.** Several formats share an extension (`.spc` , `.dat` , `.txt` , `.xml` , `.s0`). Auto-detect distinguishes them by content. Force a format only when auto-detect fails; if a forced format cannot read the file, the error message suggests returning to auto-detect.

3.3 Drag and drop, command line, multi-spectrum files

- Drop **one file** on the window to open it; drop **a folder** to open the Convert Folder dialog; drop **several files** to open the Convert Files dialog.
- `SpectraMaster.exe file1.S0` opens the file at start-up (also used by "Open with…" in Windows Explorer).
- Files holding several spectra (e.g. N42, PCF, some CNF) show a selector *Spectrum i of n* in the information panel.

3.4 Supported formats

The complete table with codes is in appendix A and in **Help › About SpectraMaster**. The most used ones in our laboratory:

Code	Format	Notes
13	Aptec v6.3 `.S0`	Read and write; counts, times, energy and FWHM calibration, Aptec sample/source records and ROIs. Aptec v3.0 (78) and v4.3 (79) are read only.
1	ORTEC CHN	MAESTRO / GammaVision; 3 calibration terms.
7	IAEA SPE	Open text format; keeps Unicode text and the calibration.
16	Canberra CAM `.CNF` (Genie-2000)	3 calibration terms; text fields are Latin-1 only (see below).
20 / 21	ANSI N42.42 XML	Recommended long-term archive format.
50	Excel `.xlsx`	Sheets *Spectrum* (channel, energy, counts) and *Info*.
60–79	Aptec import/export family	Aptec ASCII, Channel-Data CSV, Davidson, IAEA DAT, NPL, Rump, Wallac, Laval, BTI, CI-90, JCAC, ND66, Norland, Oxford, …

Calibration terms. Some formats store fewer energy-calibration coefficients than the source (e.g. CHN, SPC, CNF, S100 keep 3 terms; Laval and Norland 2). SpectraMaster then **refits** the calibration by least squares over the stored channels and warns: `… stores only 3 calibration terms: energy calibration refitted (max deviation 0.206 keV)`. Check the deviation: a few hundredths of a keV are harmless, larger values (cubic terms on 8k spectra) may shift high-energy peaks. The original coefficients are kept in the metadata (`original_energy_coeffs`) of formats that store metadata.

Arabic text in CNF, CHN and SPC. These binary formats store sample ID and description as 8-bit Latin text. Arabic letters become `?` (CNF, CHN) or are dropped (integer SPC) — without a warning. Use Latin sample IDs for files destined to Genie-2000 / MAESTRO, or keep an Aptec S0, IAEA SPE, N42, JSON or Excel copy, which preserve Arabic text.

Genie channel numbering. Genie-2000 numbers channels from 1, SpectraMaster (like ORTEC and Aptec) from 0. The calibration polynomial is shifted automatically on reading and writing CNF, so peak energies are identical in both programs; only the channel numbers differ by one.

4. Converting spectra

4.1 Save the displayed spectrum (Save As)

1. Open the spectrum (chapter 3).
2. **File › Save As…** (Ctrl + S): choose the **output format** in the list (code, name, extensions) and click OK.
3. Confirm the file name. The proposed name follows the DataMaster rule (below); the last output folder is remembered.

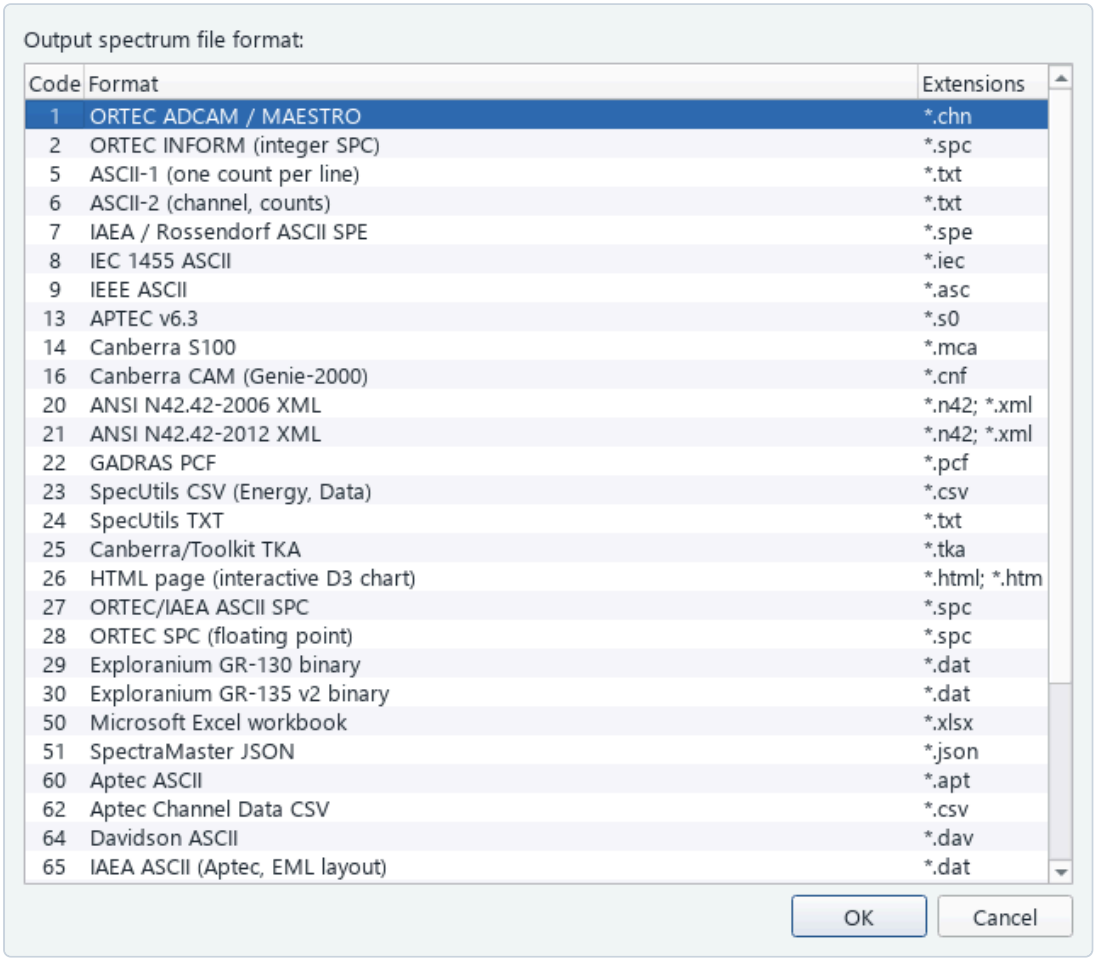


Figure 8 — Choosing the output format.

Warnings (for example a refitted calibration) appear in the status bar after saving. What a format cannot hold is never changed silently:

- **Negative counts** (after a background subtraction or Strip) are refused by the formats that store only non-negative channels (CHN, SPC, CNF, Exploranium, NPL ASCII, SPC ASCII): choose a format that keeps them, e.g. IAEA SPE, N42, JSON or Excel.
- **Fractional counts** are rounded in integer formats, with a warning giving the largest change.
- An **unknown acquisition date** is reported for formats that always store one; CHN reads its placeholder date back as unknown.
- **Arabic-Indic digits** (١٢٣) in the sample ID and description are written as 0–9 in formats that cannot hold them, with a warning.
- A spectrum with NaN or infinite counts is not written.

4.2 The DataMaster naming rule

The output name is the input name with its extension appended after an underscore, plus the extension of the output format: `CHERN.S0` → `CHERN_S0.chn`. Files with several spectra get a running number: `RUN.n42` → `RUN_n42_1.spe`, `RUN_n42_2.spe`, … This avoids overwriting when `A.S0` and `A.CHN` are both converted to SPE.

4.3 Converting many files or whole folders

- **File › Open › Select Multiple Files to Convert…** (Ctrl + M): pick any number of files.
- **File › Convert Folder…** (Ctrl + D): choose a folder, optionally **Include sub-folders**, then **Find Files**. With *All files (ignore extensions)* every file is sniffed by content.

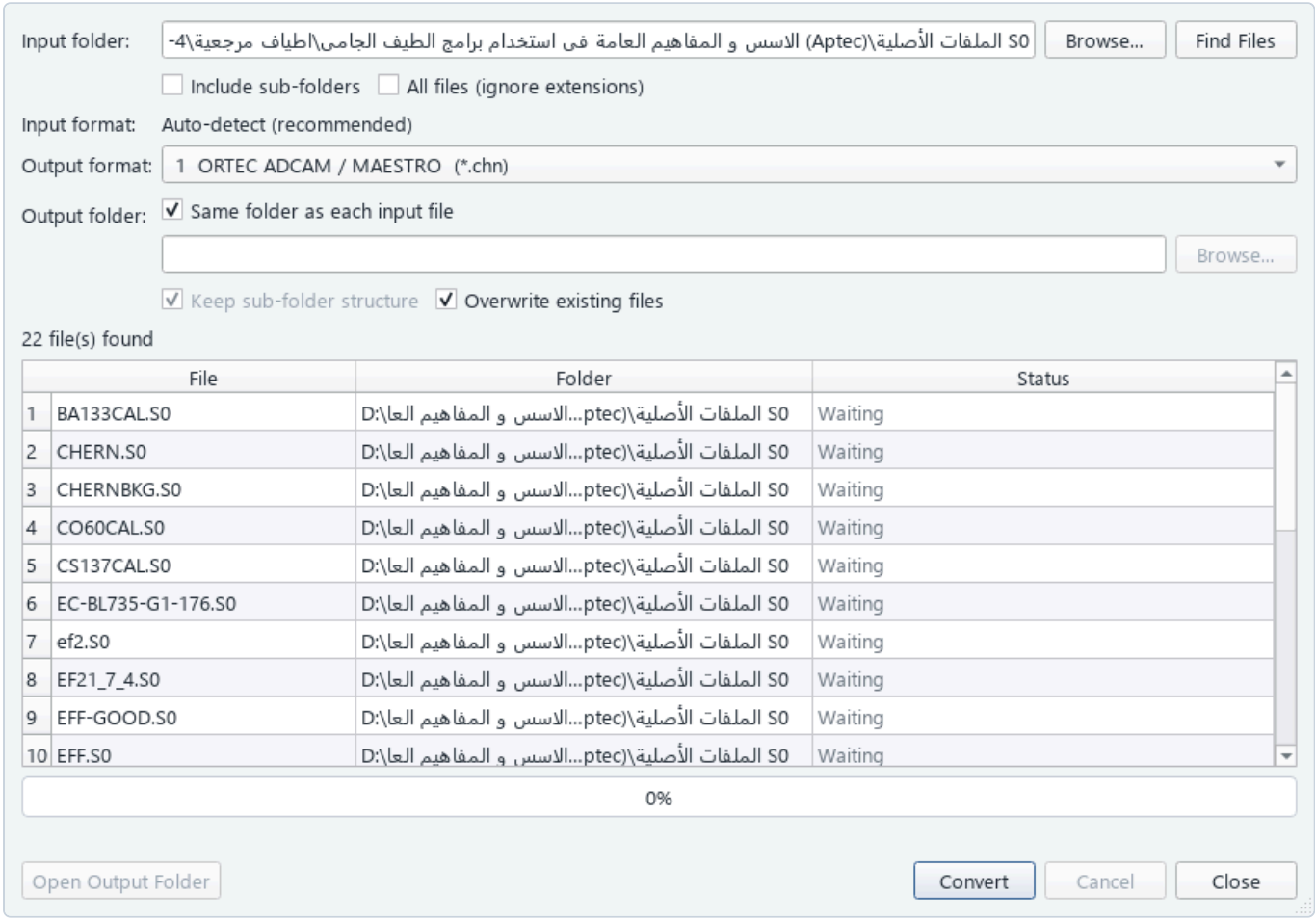


Figure 9 — Convert Folder: 22 Aptec spectra found, output to CHN next to each input file.

Choose the output format, then the output folder: **Same folder as each input file**, or one folder — with **Keep sub-folder structure** the input tree is recreated inside it. **Overwrite existing files** controls replacement. Click **Convert**; each line shows OK, the number of outputs, a warning or the error. A failure never stops the batch. **Open Output Folder** opens Explorer at the result.

> **Input format in batches.** In auto-detect mode only files whose content is recognised (or whose extension belongs to a plain-text format) are converted, so reports, Word files and pictures in the same folder are skipped. With a fixed input format the extension decides, as in DataMaster.

4.4 Command line (smconvert.exe)

DataMaster syntax

```
smconvert -i InputFile -j OutputFile          (i, j = format codes, i = 0 auto-detects)

smconvert -13 CHERN.S0 -1 "C:\Spectra\Converted Chern.chn"
smconvert -0  SAMPLE.CNF -7 SAMPLE.SPE
SpectraMaster.exe -13 CHERN.S0 -16 CHERN.CNF  (silent, no window; on error the window opens)
```

The exit code is 0 on success and 1 on failure (2 for wrong arguments), so batch files can test `%ERRORLEVEL%`. A calibration refit is printed as a warning.

Modern subcommands

```
smconvert formats                                    list all formats with codes
smconvert info CHERN.S0                              all metadata of a file
smconvert convert D:\Archive --to n42_2012 -r --out E:\N42 --keep-tree
smconvert convert *.S0 --to chn                      glob patterns are accepted
smconvert convert D:\In --from aptec_s0 --to 7 --no-overwrite -q
```

`--to` and `--from` accept the code (`7`), the key (`iaea_spe`) or an alias (`spe` , `chn` , `cnf`). Other subcommands: `peaks` , `identify` , `calibrate` , `sum` , `subtract` , `rebin` , `effcal` , `activity` , `index` , `rescue` — see the following chapters and appendix B.

> **Archive rescue.** `smconvert rescue D:\OldArchive -o E:\Rescued --to n42_2012 --to ci_cnf --resume` converts a whole archive, re-reads every output to verify counts and times, writes a log workbook and can be resumed after an interruption. The archive itself is only read. `smconvert index ROOT -o catalog.xlsx` writes a catalogue of all spectra.

4.5 GammaVision / MAESTRO integration

GammaVision and MAESTRO can call SpectraMaster as their external converter instead of DataMaster. **Help › GammaVision / MAESTRO Integration…** shows the exact **Program** and **Arguments** to enter in GammaVision (**File › Settings › Import/Export**) with **Copy** buttons.

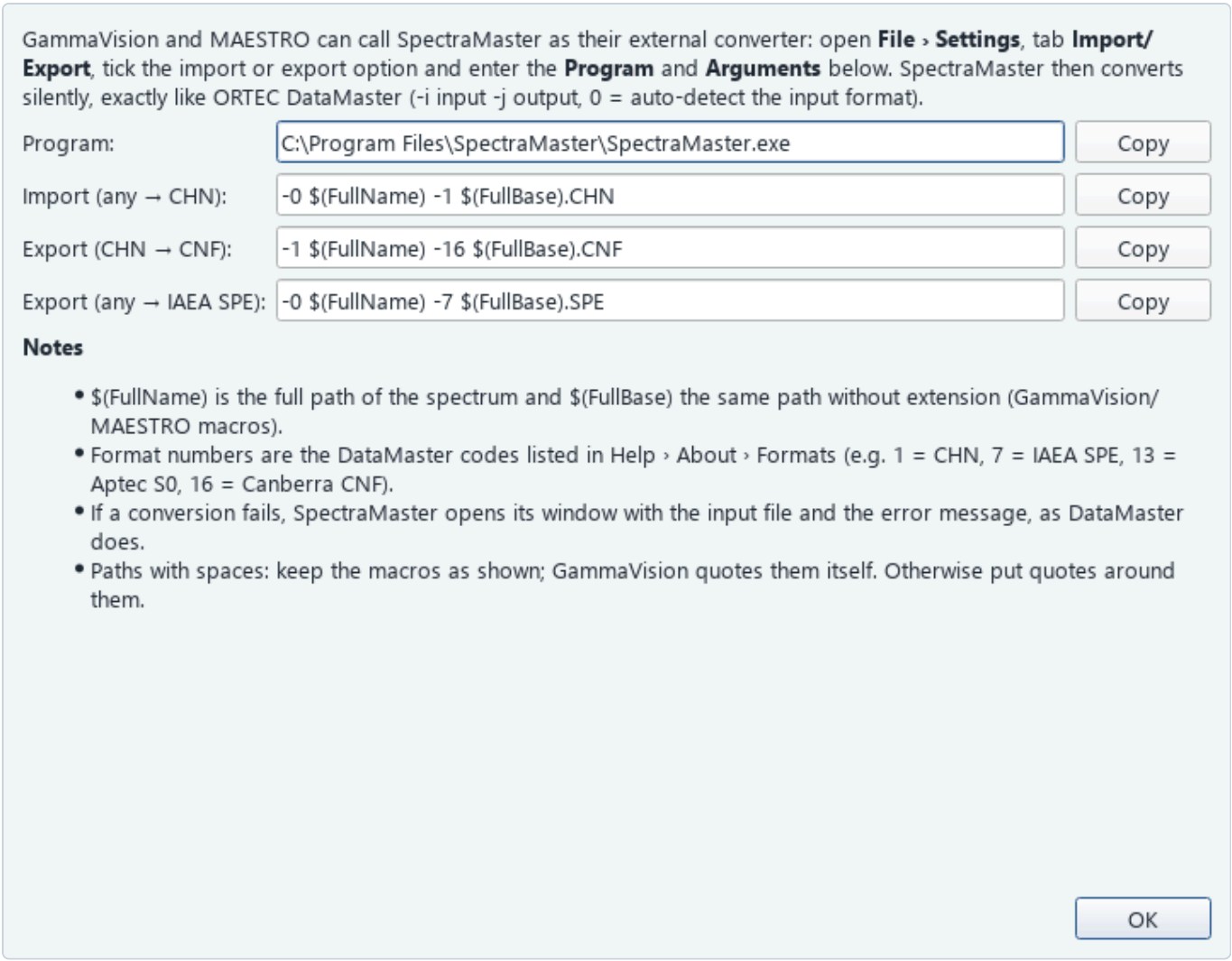


Figure 10 — Values for GammaVision / MAESTRO.

`$(FullName)` and `$(FullBase)` are GammaVision macros (full path, path without extension). If a conversion fails, SpectraMaster opens its window with the input file and the error message, as DataMaster does.

5. Viewing and zooming

5.1 Mouse and keyboard

The same list is available in the program under **Help › Mouse & Keyboard Shortcuts**.

Keys / mouse	Action
Wheel	Zoom X around the cursor
Ctrl + Wheel / ↑ ↓	Zoom Y (counts)
Shift + Wheel / ← →	Pan (Shift+← → : half a screen)
Drag horizontally	Zoom to the dragged channel range
Drag a box	Rubber-band zoom of X and Y
Ctrl + Drag	Define a region of interest (ROI)
Right / Middle drag, Shift + Drag	Pan (also Y when Y is zoomed)
Right click	Context menu
+ / −	Zoom in / out
Backspace, Alt + ←	Previous zoom
Alt + →	Next zoom
Home, Double-click, Ctrl+0	Full spectrum
Ctrl + G	Go to energy / channel
Ctrl + L	Linear / log counts axis
Ctrl + Shift + Y	Auto-scale Y to the view
Ctrl + E	Energy / channel axis
F5	Peak search & identify
Ctrl + Shift + L	Peak labels on/off
Ctrl + Shift + O	Add overlay spectrum
Ctrl + Shift + E	Export plot image (PNG / SVG)
Ctrl + Shift + C	Copy plot to clipboard
Delete (ROI table)	Delete the selected ROI
Ctrl + O	Open a spectrum (with preview)
Ctrl + M	Select multiple files to convert
Ctrl + D	Convert a folder
Ctrl + I	Open input files as (input format)
Ctrl + S, Ctrl + Shift + S	Save As (choose the output format)
Ctrl + P	Print report
Ctrl + Shift + P	Export report as PDF

Keys / mouse	Action
Ctrl + Shift + F	Archive search
F1	User manual in the interface language
Alt + F4	Exit

5.2 Zoom, history and overview strip

- **Horizontal drag** zooms to the dragged channel range (counts axis automatic); dragging a **box** zooms both axes.
- Every zoom is stored in a **history**: **Back** (Backspace) and **Forward** (Alt+→). **Reset Zoom**, Home or a double click shows the full spectrum.
- The **overview strip** under the plot always shows the whole spectrum: drag the blue window to pan, drag its edges to change the width, click elsewhere to jump, use the wheel to zoom.

5.3 Go to an energy or channel

Type an energy in the **Go to** box (Ctrl+G), e.g. `661.66`, and press Enter. The unit box selects keV or channel; the width box selects ±3/5/10 FWHM (from the FWHM calibration or the detector model) or ±5…100 keV. Uncalibrated spectra accept channels only.

5.4 Counts axis and energy axis

The counts axis can be **Linear**, **Log** or **√ (square root)** (Ctrl+L toggles linear/log). **Auto Y** fits the counts axis to the visible part. **Energy** (Ctrl+E) switches the horizontal axis between keV and channel (keV only for calibrated spectra).

5.5 Regions of interest (ROI)

Paint an ROI with F3 or Ctrl+click left then right of the peak, or let Peak Search (P) set them. The *Regions of Interest* panel lists start, end, centroid, energy, FWHM (channels and keV), gross, background and net area with ±1σ, % error, net count rate and flags, then isotope, activity and MDA after Identify and Quantify. The background follows Aptec's rules, from end points outside the ROI (ROI Properties, Shift+R). Delete with Del, **Delete** or **Clear All**; **Export CSV…** saves the table. Details in section 7.3.

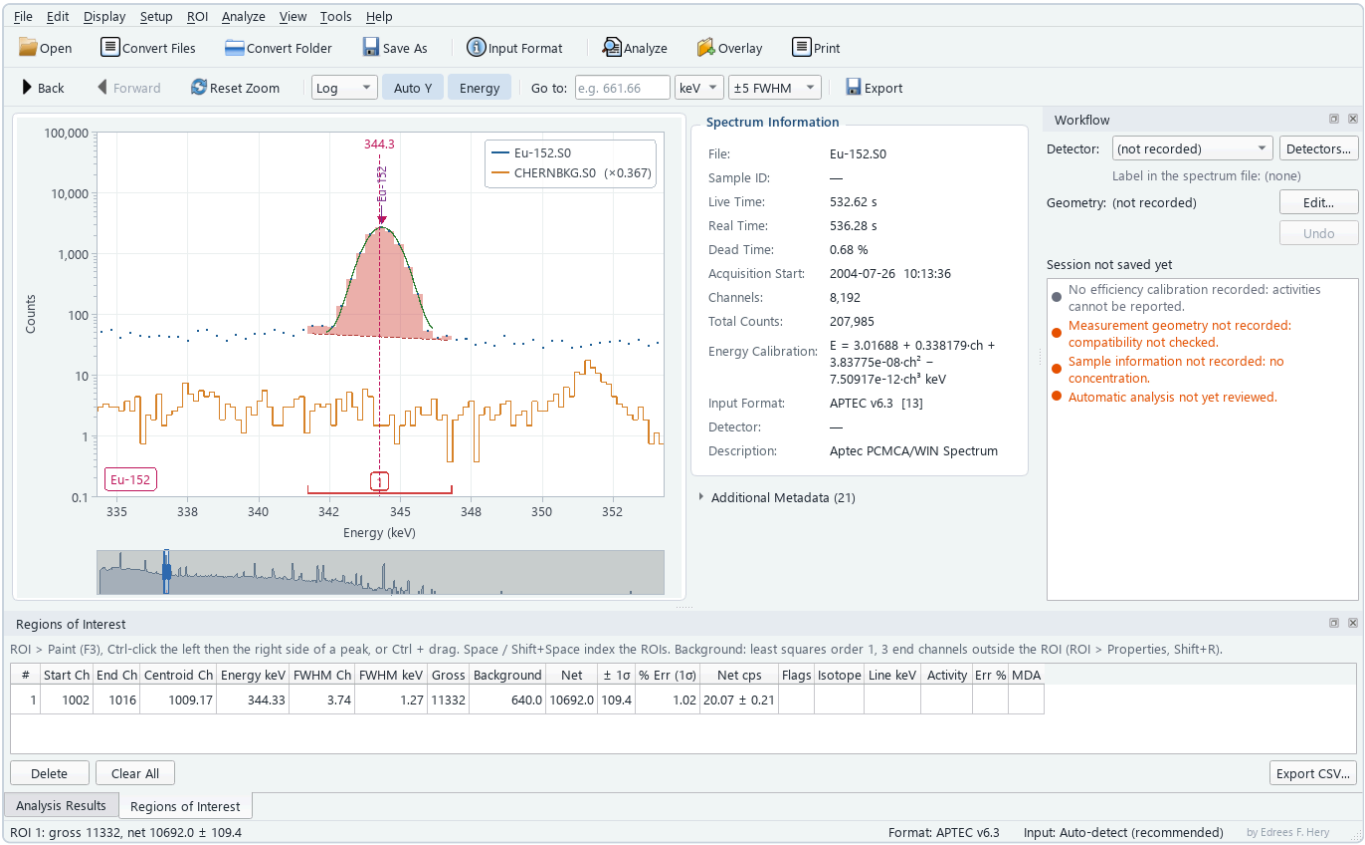


Figure 11 — Zoom on the 344.3 keV Eu-152 line with an ROI and the background spectrum CHERNBKG.S0 as a live-time normalised overlay.

5.6 Overlays

Overlay (Ctrl + Shift + O) draws other spectra — typically the background or a previous measurement — over the displayed one, each in its own colour with a legend.

View › Overlays › Normalize Overlays to Live Time scales each overlay by $LT_{displayed}/LT_{overlay}$ (the factor is shown in the legend). Overlays of different calibration are aligned in energy. Remove them one by one or all at once from the same menu.

5.7 Exporting the plot

Export (Ctrl + Shift + E) saves the current view as a PNG image or as an SVG vector drawing (for reports and publications); Ctrl + Shift + C copies it to the clipboard for Word or PowerPoint.

6. SpectraMaster analysis: fitted peaks and nuclide identification

6.1 Running the analysis

This analysis is the alternative to the Aptec method (chapter 7). Choose it in **Analyze › Setup › Analysis Method › SpectraMaster**, then press P (**Analyze › Peak Search & Identify**) or the **Analyze** button. The analysis runs in the background (a few seconds for an 8k HPGe spectrum) and opens the *Analysis Results* panel. **Analyze › Auto-analyze on Open** analyses every spectrum as soon as it is opened.

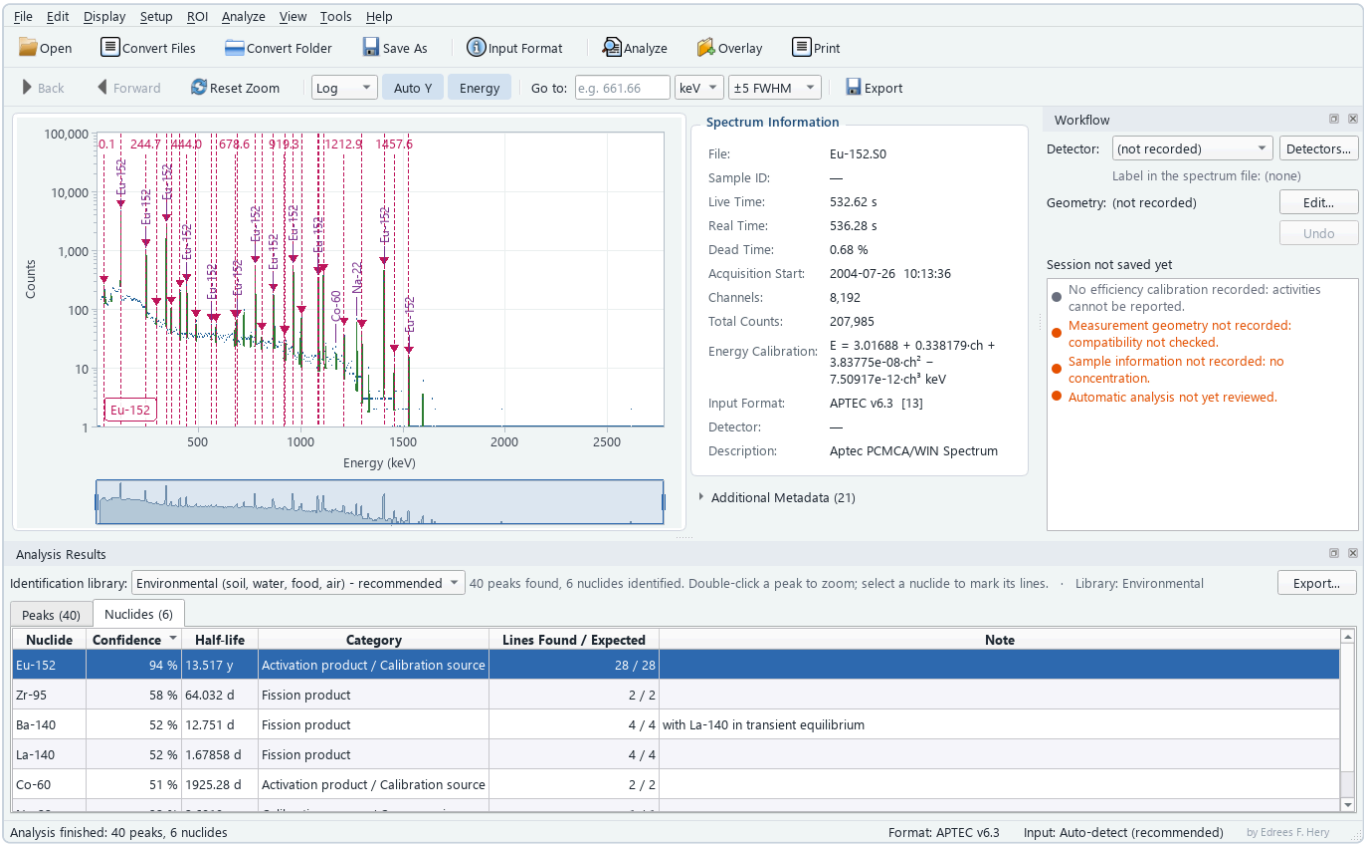


Figure 12 — Eu-152 calibration spectrum after P with the SpectraMaster method: 44 peaks, nuclides Eu-152, Eu-154 and Co-60; Eu-152 selected, its lines marked on the plot.

6.2 The peaks table

For each peak: energy (keV), centroid (channel), FWHM (keV), net area ± 1σ, count rate (cps), significance (net area / its uncertainty) and the assigned nuclide. A double click zooms to the peak. Notes mark special peaks: *X-ray region*, *annihilation* (511 keV), *single / double escape*, *sum peak* (e.g. 1173+1332), *possible backscatter*.

How it works, in brief: the FWHM model comes from the FWHM calibration (or the energy calibration and a detector-type guess) and is rescaled to the strongest isolated peaks; for HPGe without a FWHM calibration its energy dependence is corrected too, between the lowest and highest of those peaks. Peaks are searched with a Gaussian matched filter (second-difference method) and accepted above a significance of 3; neighbouring peaks are fitted together as multiplets (Gaussians + background + step), and areas and uncertainties come from the fit covariance. HPGe peaks sit on a straight background; scintillator (NaI) peaks on a log-quadratic

continuum whose curvature is measured on a wider region that stops before the neighbouring peaks. When the detector's peak shape was measured (section 6.6), peaks are fitted with its low-energy tail.

6.3 The nuclides table and the confidence value

Each candidate nuclide shows its **confidence**, half-life, category (natural, fission, activation, calibration, medical, industrial, actinide; nuclides with two common origins show both, e.g. Cs-137 "fission / calibration"), the number of lines found / expected and a note (series membership, "typical background", interferences such as U-235 185.7 keV under Ra-226 186.2 keV). Selecting a nuclide marks all its library lines on the plot; peaks that match are highlighted.

> **The confidence is a heuristic score (0–100 %), not a probability.** It combines how much of the expected (detectable) line pattern is explained, the energy agreement, the statistical significance, penalties for a missing strongest line or inconsistent area ratios, and a greedy treatment of shared peaks. Weak lines below the local detection limit are not counted as missing. Results — especially low-confidence ones, single-line nuclides and short-lived nuclides in old spectra — **must be reviewed by an expert**. The identification does not replace the laboratory's validated analysis procedure.

Peak labels (Ctrl + Shift + L) write the nuclide names on the plot; **Export…** in the results panel saves peaks, nuclides and spectrum details to Excel or CSV.

6.4 Uncalibrated spectra

Without an energy calibration peaks are listed by channel only and no nuclide is identified (a message explains this). Calibrate the spectrum first (chapter 8), e.g. with Eu-152 or Co-60/Cs-137 lines.

6.5 The nuclide library

The built-in library contains 200 nuclides with 2,732 gamma and X-ray lines. Values are taken from the IAEA ENSDF evaluation (IAEA *ndlab* database behind LiveChart of Nuclides) and cross-checked against Sandia National Laboratories' *SandiaDecay*. Short-lived daughters in equilibrium (Ba-137m, Pr-144, Rh-106, La-140 …) are included in the parent's intensities; natural-series members are separate entries. On the command line another library can be used: `smconvert identify FILE --lib mylib.csv` with lines `nuclide, energy_keV, intensity_%[, half_life, category]`.

Limitations

- The identification does not correct true-coincidence summing and uses no detailed efficiency model: the expected line ratios use a generic HPGe/NaI efficiency shape. (The activity calculation can correct summing, section 9.5.)
- Strongly overlapping multiplets and very low-statistics spectra can give false or missed identifications.
- Decay since the measurement is not considered: a nuclide is proposed if its lines are present, even if it is implausible for the sample history.

6.6 Detector profiles, peak shape and analysis sessions

The **Workflow** panel (**View › Workflow Status**) shows the analysis session of the displayed spectrum: its detector, geometry, calibrations and the status of each step. **Detectors…** opens the detector profiles. A profile

keeps a permanent id, so renaming a detector keeps its calibrations and sessions. Choosing the detector of a measurement is an undoable edit, and an efficiency calibration of another detector is then reported as incompatible.

Peak Shape… in the detector profiles measures the low-energy tail of that detector's peaks from calibration spectra with strong, well separated peaks. You confirm it before it is recorded. From then on, the analysis of that detector's samples fits peaks with this shape. Without it, peaks are Gaussian.

File › Save Session stores the session (`.smsession`): the spectrum reference with its SHA-256 fingerprint, the calibrations, the background and reagent blank files, the results and the edit history. When a session is opened, the spectrum file is checked against the fingerprint. Results computed by an older analysis engine are shown as stored and marked as requiring recalculation.

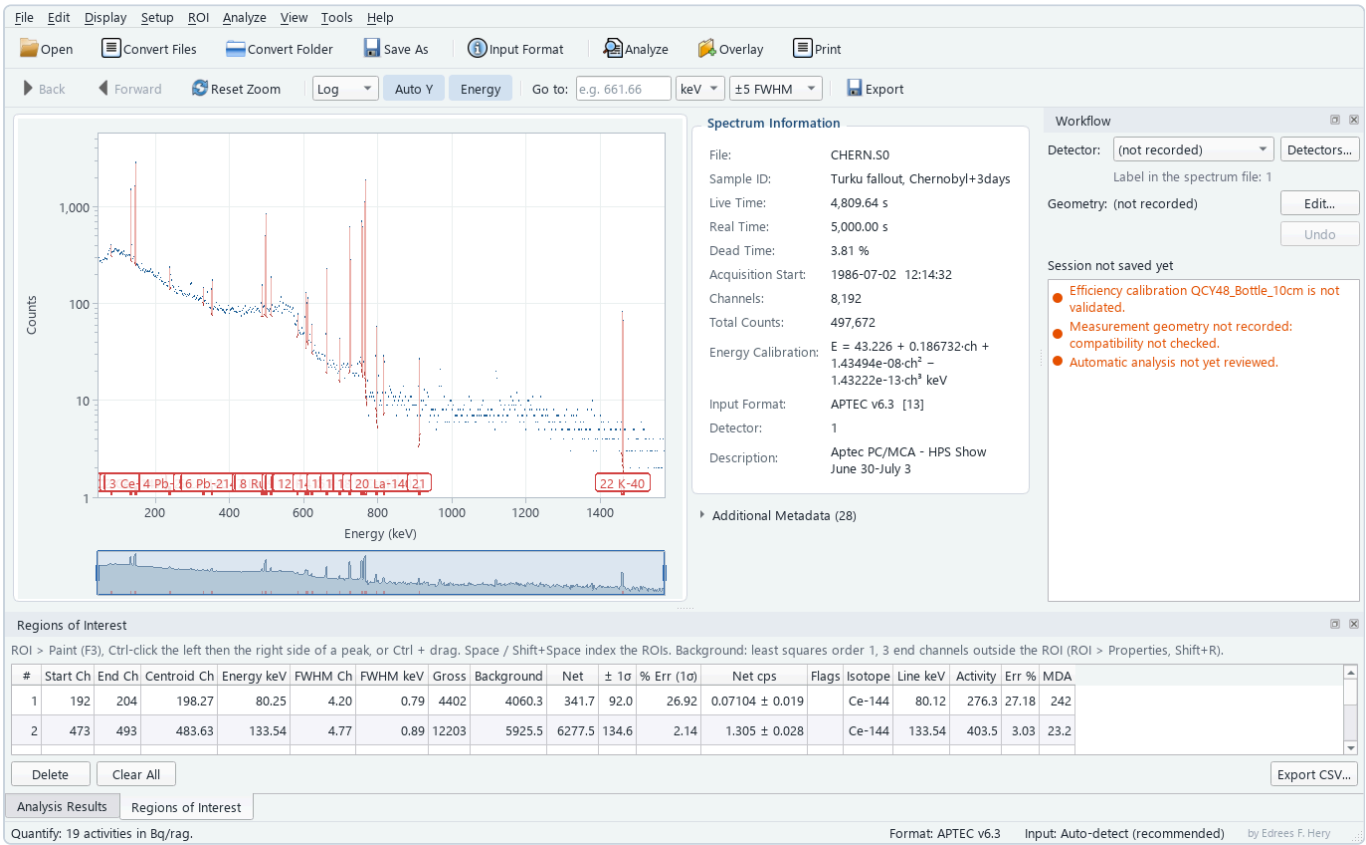
7. The Aptec method: menus and workflow

SpectraMaster follows the menus, keys and workflow of **Aptec PCMCA/WIN 6.31**, so that Aptec users need nothing new. The menu bar, in order: **File, Edit, Display, Setup, ROI, Analyze, View, Tools, Help**. The method works on **regions of interest (ROIs)**: ROIs are set on the peaks, identified, and quantified. **The Aptec method is the default**; the fitted-peak analysis of chapter 6 is chosen in **Analyze › Setup › Analysis Method**.

Aptec's algorithms were implemented as its help file describes them and compared with the results Aptec itself stored in its spectrum files (section 7.8). Aptec was not decompiled and none of its code was copied.

7.1 The workflow at a glance

1. Calibrate	F2 quick energy calibration, or Shift + F2 full energy, resolution and efficiency calibration (section 7.2).
2. Describe	F5 Sample Description, F6 Quantity Description (quantity, units, activity time).
3. ROIs	P Peak Search sets the ROIs, or F3 / Ctrl +click paints them.
4. Multiplets	M Multifit decomposes multiplets into Gaussian peaks.
5. Identify	I identifies the ROIs with the isotope library.
6. Background	B subtracts the background count rate of identified ROIs (optional).
7. Activity	Q Quantify: activity or concentration and MDA of every ROI.
8. Results	The *Regions of Interest* list, CSV export and reports (chapter 10).



#	Start Ch	End Ch	Centroid Ch	Energy keV	FWHM Ch	FWHM keV	Gross	Background	Net	± 1σ	% Err (1σ)	Net cps	Flags	Isotope	Line keV	Activity	Err %	MDA
1	192	204	198.27	80.25	4.20	0.79	4402	4060.3	341.7	92.0	26.92	0.07104 ± 0.019		Ce-144	80.12	276.3	27.18	242
2	473	493	483.63	133.54	4.77	0.89	12203	5925.5	6277.5	134.6	2.14	1.305 ± 0.028		Ce-144	133.54	403.5	3.03	23.2

Figure 13 — A soil spectrum (CHERN.S0) after P, I with Aptec's isotope.lib, and Q: isotope names on the ROIs, activities and MDAs in the ROI list.

7.2 Setup menu: calibrations

- **Quick Energy Calibration** (F2): up to 16 channel ↔ energy points; put the channel marker on a peak and type its energy. A first- or second-order polynomial is chosen from the number of points.
- **Full calibration** (Shift+F2, S_Calib style): energy and resolution (FWHM) with up to 256 points and degree 5, and efficiency with Aptec's types (5th or 3rd order ln-ln, polynomial, cubic spline). Points are loaded from the ROIs (*Load ROIs*), activities from a certificate (**Edit** › **Certificate**), decay-corrected to the measurement date.

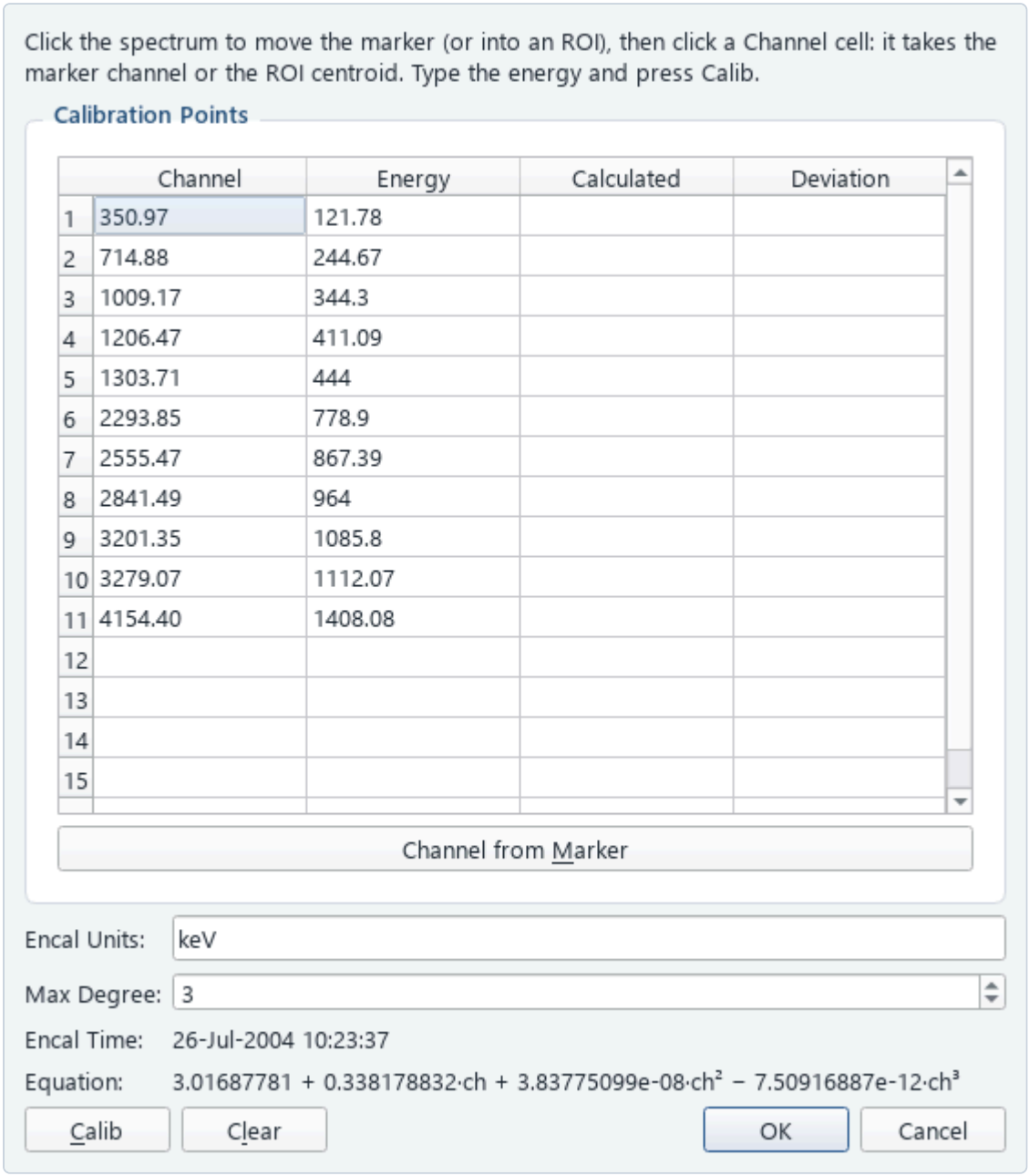


Figure 14 — Quick Energy Calibration (F2).

7.3 ROI menu

- **Properties** (Shift+R): background line (straight line between the end-point averages, or least squares of order 1–5), number of end points (fixed, or varying from a value at channel 0 to a value at the last channel), centroid range and weighting, MDA sigma (default 1.645) and error sigma.

- **Paint** (F3, or Ctrl+click left then right of the peak): the Paint bar at the bottom expands, shifts, sets (Enter) or cancels (Esc) the ROI; its statistics are shown while painting.
- **Edit** (F4) the indexed ROI, **Delete Indexed** (Del), **Delete All**, **Sort**, **Flag Definitions**. Space / Shift+Space index the next / previous ROI.

ROI statistics follow Aptec's rules: end points outside the ROI, touching ROIs share one background, net = gross − background, % error (1σ) = √(gross + background) / net × 100 and MDA = (K^2 + 2K√(2B)) / live time. They reproduce the values Aptec stored for 73 of 79 ROIs.

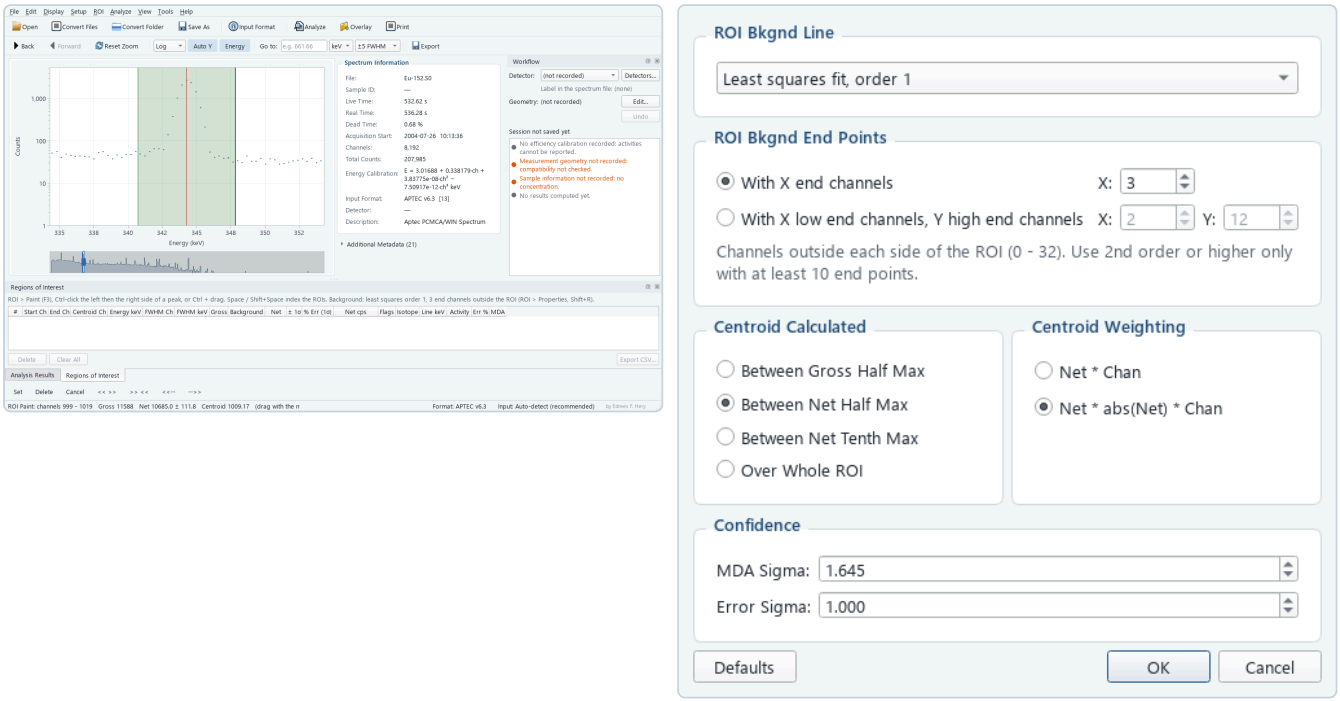


Figure 15 — Painting an ROI on the 344.3 keV line with the Paint bar, and ROI Properties.

7.4 Analyze menu

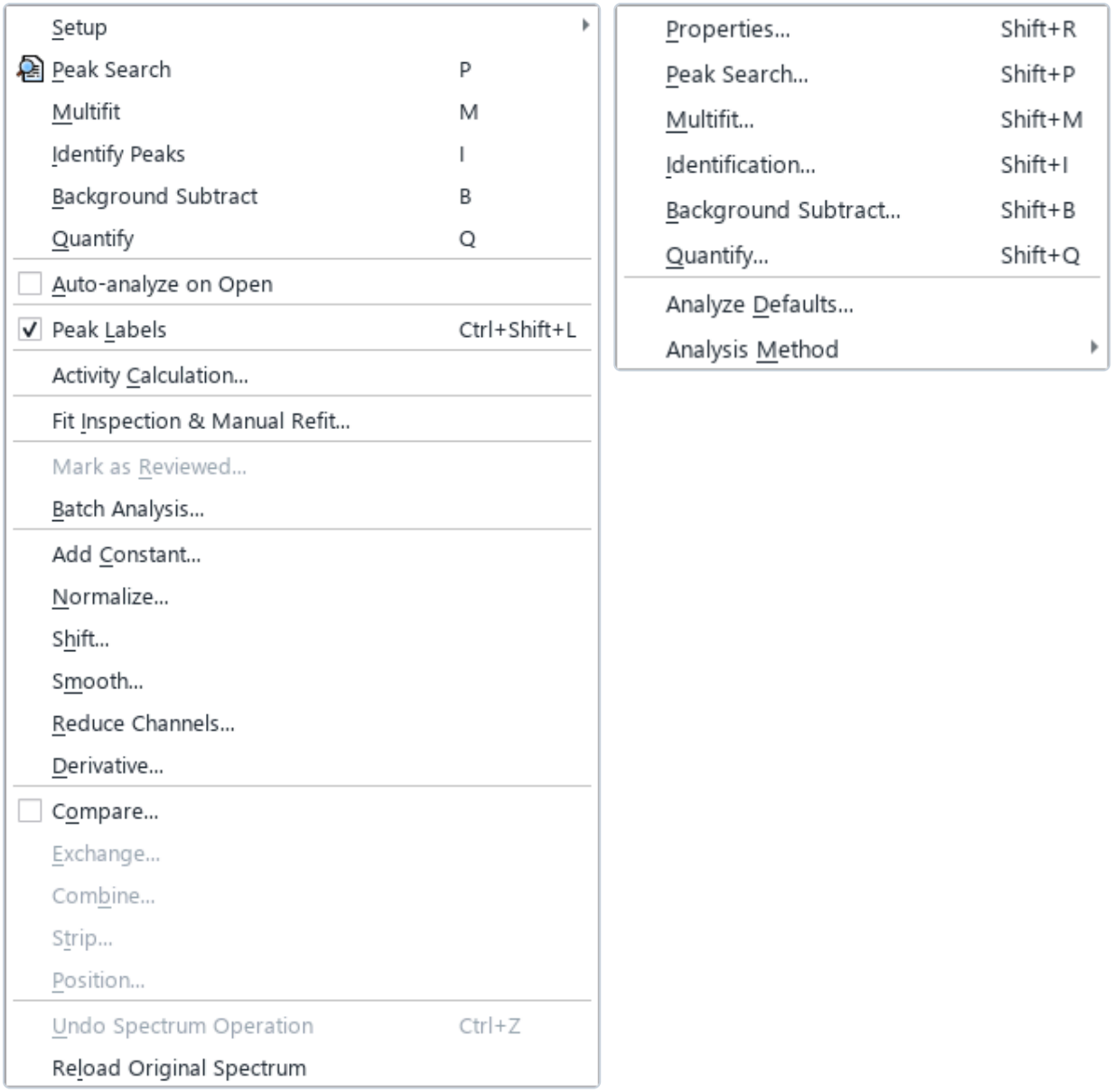


Figure 16 — The Analyze menu in Aptec's order and its Setup submenu.

Setup ›

ROI Properties (Shift + R), Peak Search (Shift + P), Multifit (Shift + M), Identification (Shift + I), Background Subtract (Shift + B), Quantify (Shift + Q), **Analyze Defaults** (factory defaults, or save and recall your own defaults for all of them at once) and **Analysis Method** (Aptec or SpectraMaster).

Peak Search (P)

Sets ROIs on the peaks, rejects every ROI whose 1-sigma net error is above the **Max Peak Error** and recalculates the backgrounds until the result is stable; the search can be limited to a channel range (*Partial*). **Note:** the search follows the Aptec help but does not reproduce Aptec's ROIs exactly: it finds about 79 % of them (section 7.8). Always review the ROIs and add or delete with F3 and Del.

Multifit (M)

Decomposes each ROI into Gaussian peaks by Marquardt-Levenberg least squares. An ROI is a multiplet when it is wider than the resolution calibration expects or shows several significant centroids. The width is fixed from

the resolution calibration (or varying) and varies automatically when 511 keV lies in the ROI; the background is fixed from the end points (or varying). A peak is kept when its amplitude ≥ K × its uncertainty (K = 1.645). Without a resolution calibration one is made from the ROI widths. Each fitted peak becomes its own ROI, flagged `m` (a fitted singlet `s`), so Identify and Quantify use the separated peaks; each Gaussian and their sum are drawn. Changing the ROI Properties removes the fits.

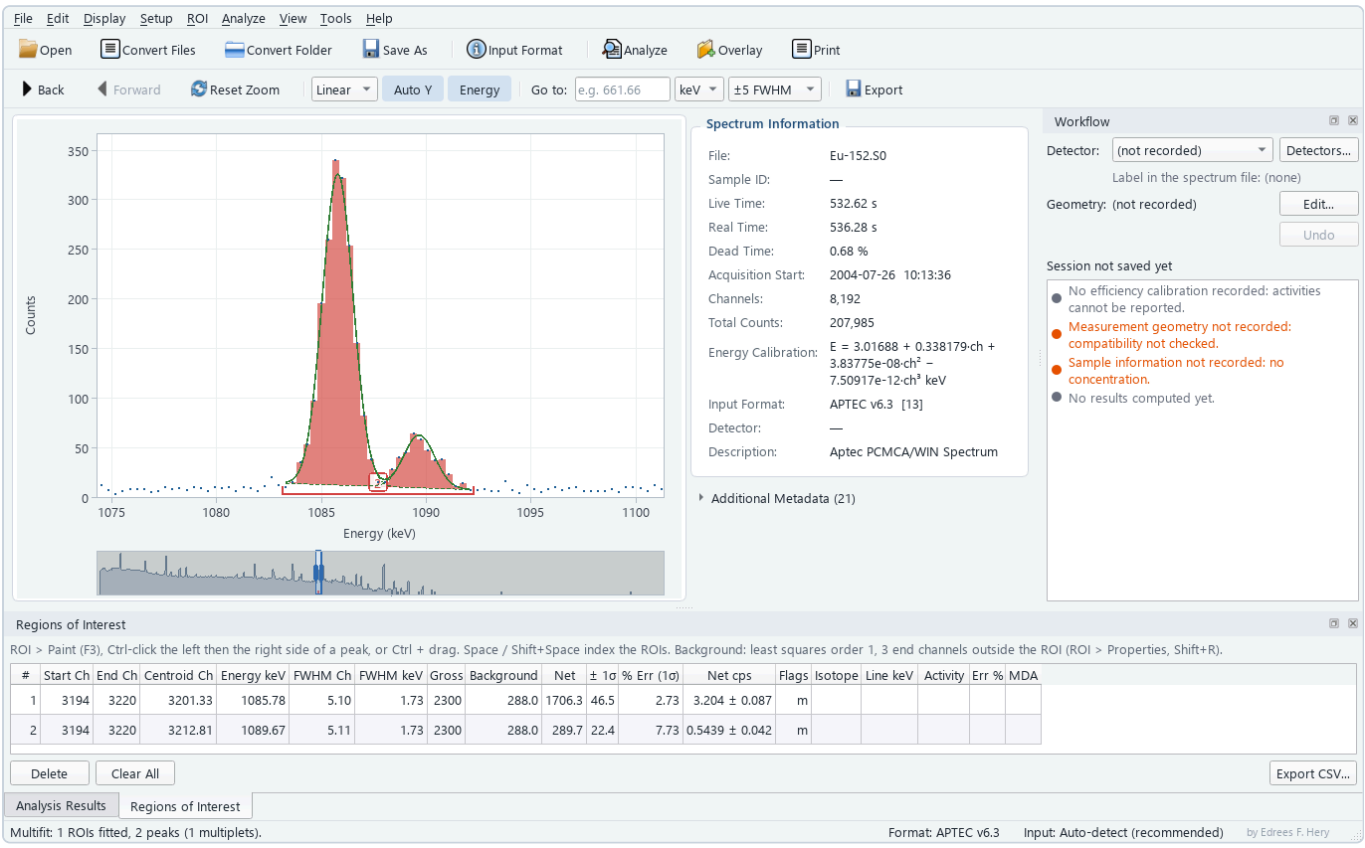


Figure 17 — The Eu-152 doublet 1085.8 / 1089.7 keV after `M`: two Gaussians (dashed) and their sum.

Identify Peaks (`I`)

With an Aptec library (`isotope.lib`) or SpectraMaster's bundled library. Library lines within the **tolerance** of the ROI centroid energy are candidates: a fraction of the ROI FWHM (default 0.5) or a fixed tolerance in keV. An isotope is accepted when its lines with an ROI carry at least the **Fraction Limit** (%) of the yield of all its lines inside the spectrum's energy range. Isotopes decayed beyond the **Half Life Limit** can be rejected.

Background Subtract (`B`)

Choose the background spectrum in `Shift` + `B`. It is searched and identified with the same setups, and the net count rate of the same isotope line is subtracted when it is above its MDA; uncertainties are combined in quadrature (flag `b`). No background-subtracted Aptec file was available to compare with.

Quantify (`Q`)

Activity = net rate / (efficiency × yield), decay-corrected from the count start to the **activity time** (`F6`) and converted to the chosen units; as **activity** in the sample, **concentration** (divided by the quantity) or **total activity** (concentration × total quantity), multiplied by the W/X/Y constants when selected. Emission units (gamma/s) are not divided by the yield. The efficiency comes from the spectrum's efficiency calibration or a saved curve, **at the ROI centroid energy** as Aptec does.

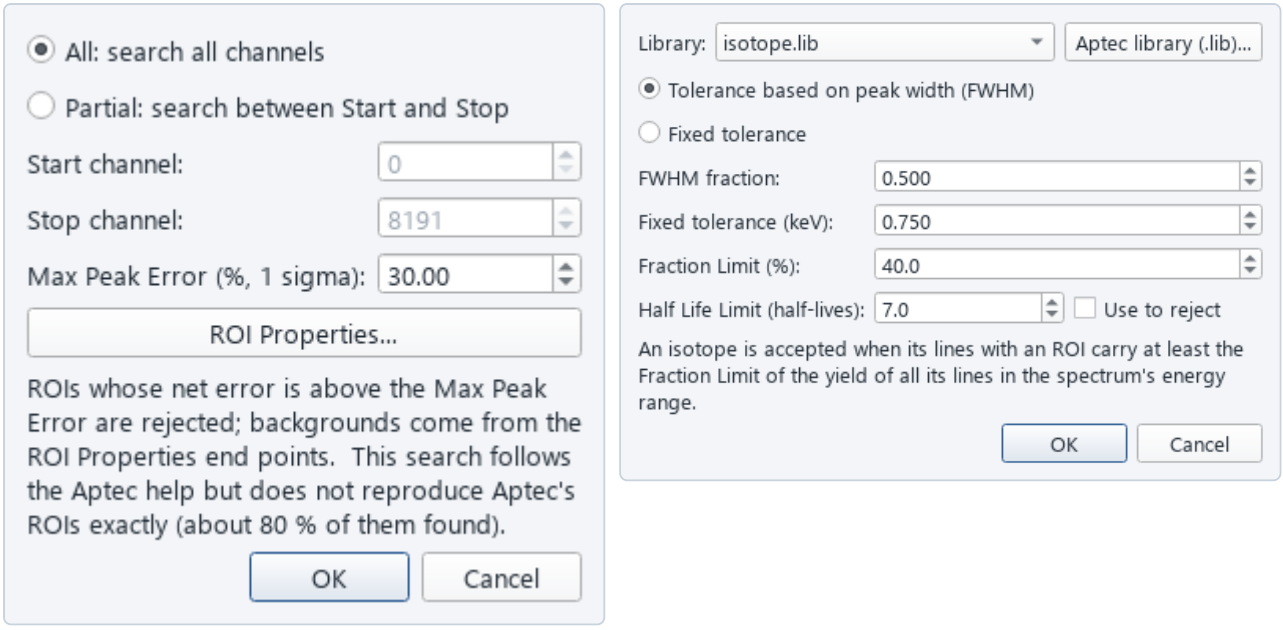


Figure 18 — Peak Search setup (Shift+P) and Identification setup (Shift+I).

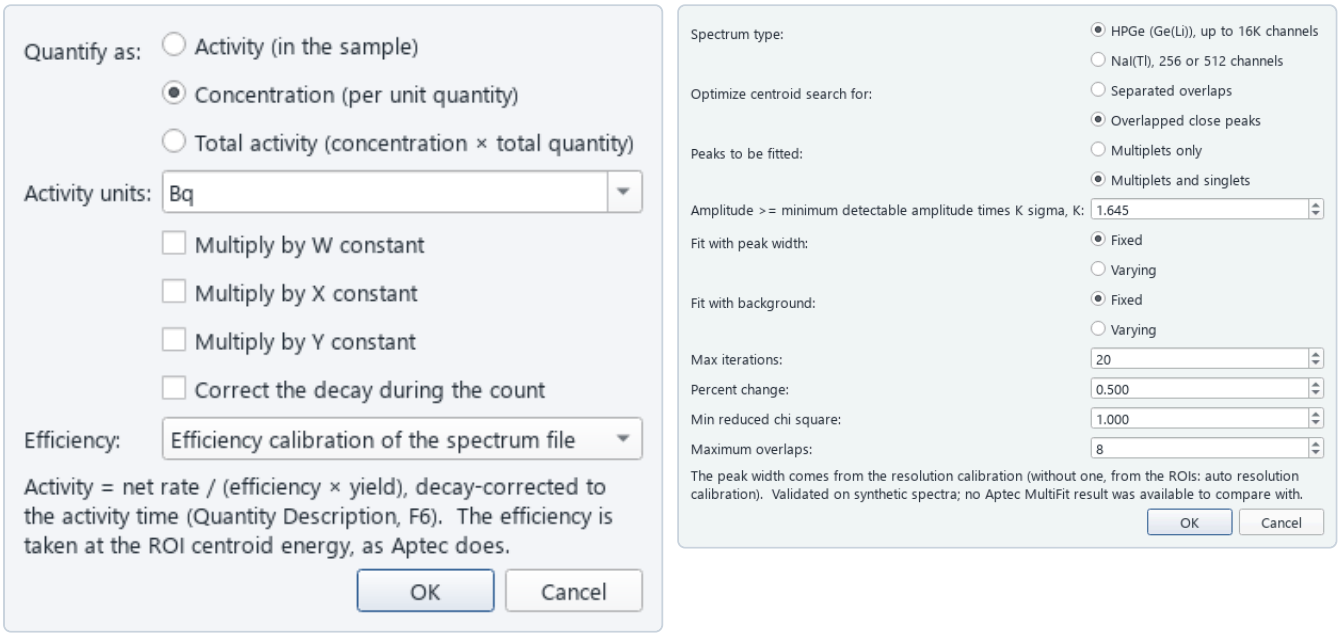


Figure 19 — Quantify setup (Shift+Q) and Multifit setup (Shift+M).

Results in the ROI list

The *Regions of Interest* list gains **Isotope**, **Line keV**, **Activity**, **Err %** and **MDA**, all exported to CSV. Aptec's flags in the Flags column:

`?`	Net rate (or activity) below the MDA.
`m` / `s`	Peak of a fitted multiplet / fitted singlet (Multifit).
`b`	Background subtracted.
`t`	Not decay-corrected over the count (count longer than 7 half-lives).
`>`	Decay correction to the activity time overflowed.

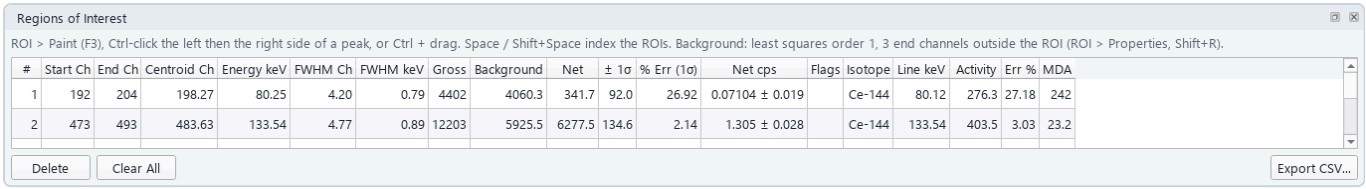


Figure 20 — The ROI list after Identify and Quantify.

7.5 Spectrum operations (lower Analyze menu)

- **Add Constant**, **Normalize** (counts and live / true times; with a compare spectrum the live-time ratio is proposed), **Shift** by whole channels, **Smooth** over 3, 5, 7 or 9 points, **Reduce Channels** (calibrations and ROIs follow), **Derivative** 1st / 2nd / 3rd ("Original data" returns).
- **Compare** another spectrum file over the displayed one, **Exchange** them, **Combine** / **Strip** the compare spectrum channel by channel (after a confirmation showing both live times) and **Position** it vertically (also Ctrl + ↑ / ↓).
- **Undo Spectrum Operation** (Ctrl + Z, 20 steps) and **Reload Original Spectrum**.

> The operations change only the spectrum on the screen; the file is unchanged until you save it (**File › Save As**). After smoothing, stripping, a derivative or a normalization the counts are no longer Poisson counts and uncertainties from √counts are not valid (the status bar warns). Aptec does not document its smoothing weights (binomial here) or derivative formula, so these were not matched to Aptec.

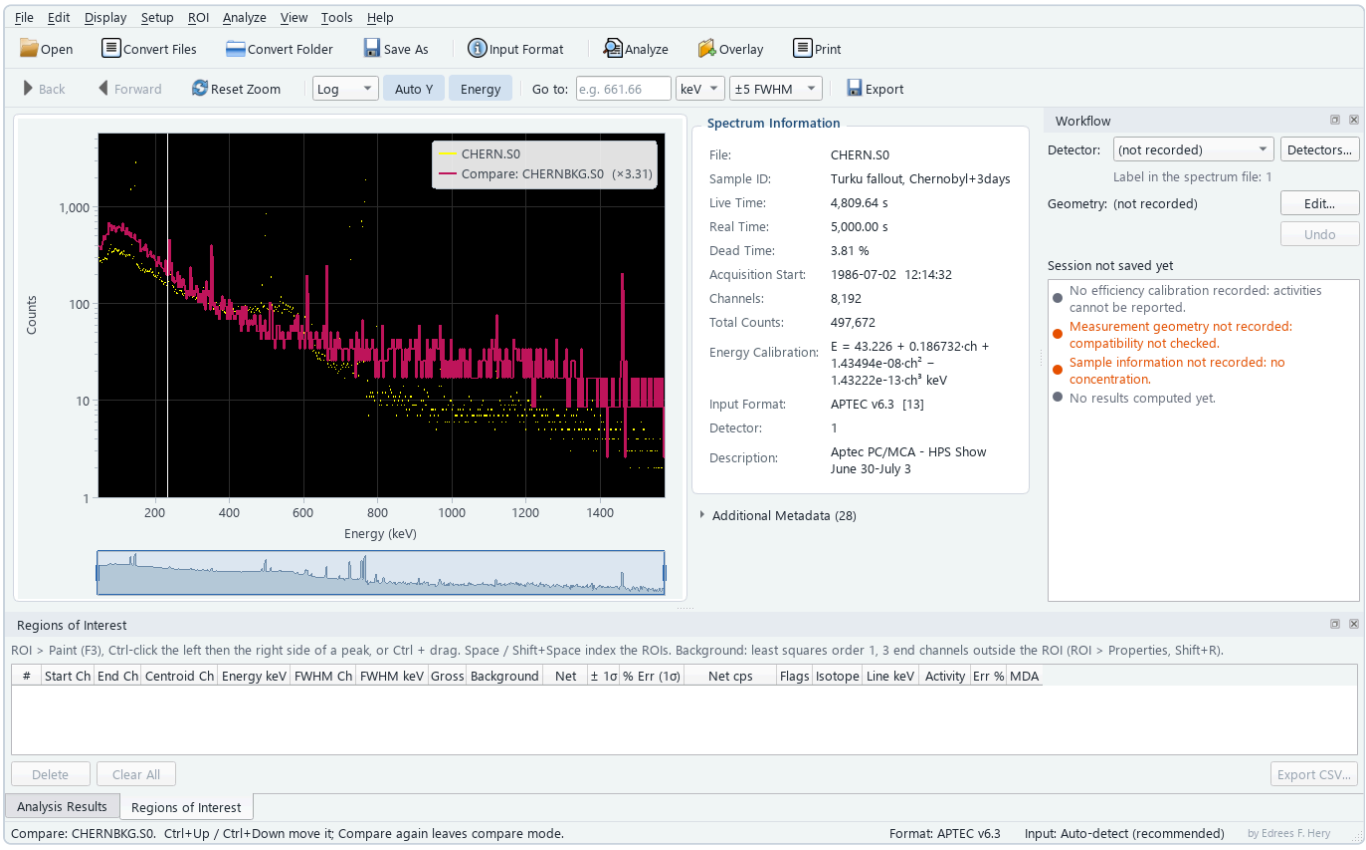


Figure 21 — Compare mode: the background spectrum CHERNBKG over the sample spectrum, with Aptec's colours (yellow on black).

7.6 View menu

Preferences, **Color** (default, black & white for printing, or **Aptec colors**), the **QA Control Bar** (Ctrl + 8), **Default Position**, **Save Settings on Exit** and the **split display**, horizontal (H) or vertical (V). The spectrum is drawn as **points** as in Aptec; toggles: Plot All Points (L), Fill Spc (F), Grid (G), Connect Points (N), Show Fitted ROIs (K), Fill ROIs (Shift + F), ROI Data box (D), ROI Display (Shift + D), ROI Labels (J), Channel Marker (Shift + J), FWHM Percent (%). Vertical scale: auto (A), auto zero (Z), next / previous type (S / Shift + S); horizontal: channels (C) or energy (E). The arrow keys move the channel marker, PgUp / PgDn the spectrum window, Esc shows the whole spectrum. All keys: section 5.1.

7.7 Edit and Display menus

- **Edit**: Select All / Select Region of the display and Copy to ClipBrd as a picture (Ctrl + Ins, clear with Ctrl + Del); **Sample Description** (F5: title, IDs, sampling times); **Quantity Description** (F6: quantity and units, W/X/Y constants, activity units and time, total quantity); **Spectrum Data** (correct the acquisition times, with care); **Certificate** (calibration-source activities); **Library**; **Decay Correction Calculation**; **Load Header Information** from another spectrum file (calibrations, descriptions, ROIs); **Reset Header**.
- **Display**: the normal spectrum, or the energy, resolution or efficiency calibration with its points (Space indexes the points and shows the calculated value and deviation).

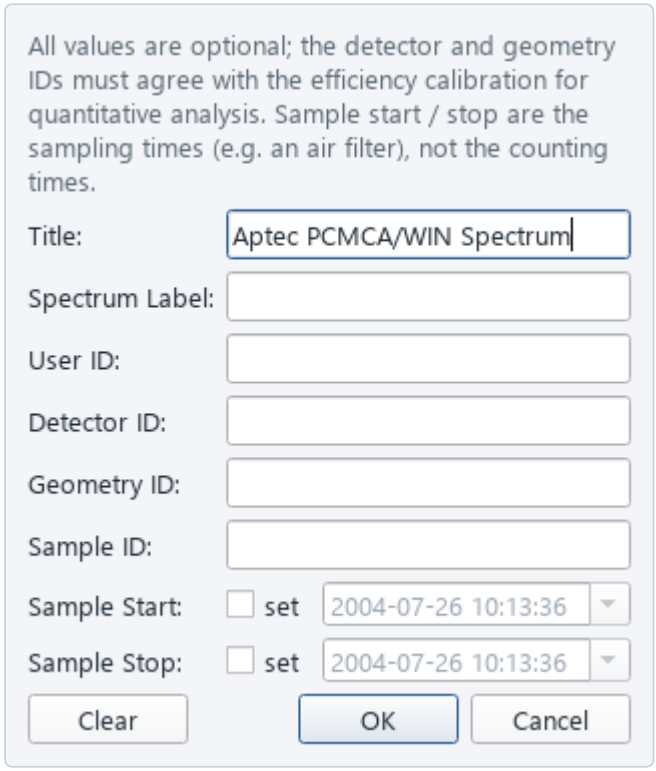


Figure 22 — Sample Description (F5).

7.8 Comparison with Aptec

Each step was compared with the results Aptec stored in its spectrum files (the read-only laboratory archive; only numbers were published), with success limits set **before** measuring and evaluated once on a separate test set:

Step	Result	Verdict
ROI statistics	gross 79/79; background, net, error 73/79	agree
Peak Search (4203 spectra)	finds 79 % of Aptec's ROIs, precision 62 %, edges 2 channels apart	**fail** (limits 90 %, 85 %, 1 channel)
Identify (736 ROIs)	agrees with Aptec for 88 %	**fail** (limit 95 %)
Quantify (49 ROIs)	49/49 within 0.003 % once the efficiency is taken at the centroid energy as Aptec does (45/49 with the pre-registered method)	agree (correction found afterwards)
Multifit	no Multifit result in any file; synthetic validation: bias ≤ 2.8 % for doublets ≥ 1 FWHM apart	pass (synthetic)
Background Subtract, smoothing, derivative	no Aptec data to compare with	not verified

Details in `docs/baseline/aptec_*` of the program's repository. Agreement with Aptec does not prove either program correct and does not replace the laboratory's validation procedures.

8. Energy recalibration and spectrum arithmetic

These tools are available on the command line (`smconvert`); the results are ordinary spectrum files that can be opened in the program. The input files are never modified.

8.1 Energy (and FWHM) recalibration

```
smconvert calibrate Eu-152.S0 --source Eu-152                propose a calibration from source lines
smconvert calibrate *.S0 --source Co-60,Cs-137 -o D:\Recal    write recalibrated copies
smconvert calibrate X.spe --pairs 1893.7:661.66,3364:1173.23,3812:1332.49 --order 2 -o D:\Recal
smconvert calibrate X.spe --coeffs 0.12,0.3501,1.2e-8 -o D:\Recal  apply known coefficients
smconvert calibrate *.chn --source Eu-152 --only-uncalibrated -o D:\Recal
```

`--source` finds the peaks of the given nuclide(s) (using the existing calibration as a guide when present), fits an energy polynomial (order 1–3) and, from the peak widths, a FWHM calibration; it prints a table with channel, given and fitted energy and residuals (RMS and maximum). Without `-o` nothing is written. `--fwhm ch:keV,…` fits a FWHM calibration from given pairs.

```
Energy calibration, order 2: c0=2.85491689  c1=0.33842267  c2=-4.33136557e-08
  #     channel       given      fitted   residual
  1     351.149     121.782     121.686    +0.0957  Eu-152 121.78
  ...
RMS residual 0.0615 keV, max |residual| 0.1460 keV, 8 degree(s) of freedom
```

8.2 Sum of spectra

```
smconvert sum A.S0 B.S0 C.S0 -o SUM.spe [--align auto|channel|energy]
```

Counts, live and real times are added. With different calibrations the spectra are aligned in energy (rebinned onto the first spectrum's calibration); `--align channel` forces a channel-by-channel sum.

8.3 Background subtraction

```
smconvert subtract CHERN.S0 CHERNBKG.S0 -o CHERN_net.spe [--scale K] [--clip] [--sigma-csv net.csv]
net = sample - 3.31379 x background; net counts 301801 +- 1071
```

The background is scaled by $LT_{sample}/LT_{background}$ unless `--scale` is given. `--clip` sets negative channels to zero; `--sigma-csv` also writes channel, energy, net counts and the propagated 1σ per channel.

8.4 Rebinning and truncation

```
smconvert rebin CHERN.S0 --factor 2 -o CHERN_4k.spe          8192 -> 4096 channels
smconvert rebin X.S0 --start 100 --stop 4196 --renumber -o part.spe
```

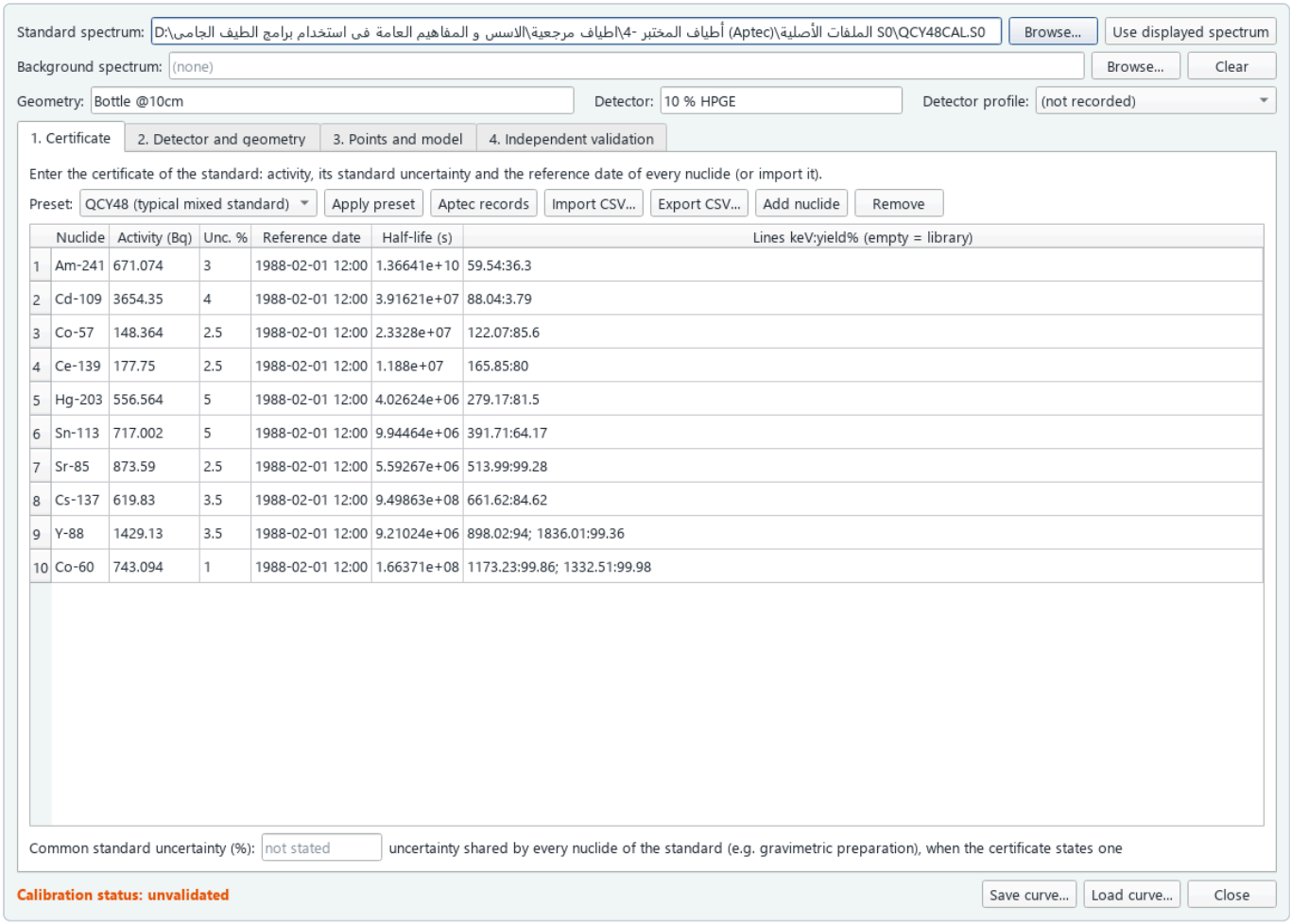
The calibration is transformed so that energies stay correct.

9. Efficiency calibration and activity

> **Before you start.** The efficiency curve is valid only for the **same detector, geometry (container, filling height, distance) and matrix density** as the standard. True-coincidence summing (TCS) is corrected only when you enable it and give the peak-to-total ratio of your detector and geometry (section 9.5); otherwise cascade emitters such as Co-60, Cs-134, Y-88 and Eu-152 are flagged. Self-attenuation is corrected only by a **manual factor**. Results must be validated by the laboratory.

9.1 Efficiency calibration

1. Open the standard spectrum (here `QCY48CAL.S0`, 10 ml bottle at 10 cm, HPGe) and choose **Analyze › Efficiency Calibration…**. The displayed spectrum becomes the standard (or use **Browse…**).
2. Fill the **certificate**: one row per nuclide with activity (Bq), uncertainty %, reference date, half-life (s) and optionally the lines `keV:yield%` (empty = library values). Sources: **Aptec records** (the source records stored in Aptec `.S0` files), a **Preset** (QCY48, QCRB1241, Eu-152 point source), **Import CSV…** or typing.
3. Choose the **model**: *two-branch log-log* (recommended for HPGe: low- and high-energy branches joined at a crossover energy, automatic or fixed) or a single *log-log polynomial* of order 1–5. Enter geometry and detector names.
4. Click **Measure & Fit**. Each line becomes a point: net area (with background-peak subtraction if a background spectrum is used in the CLI), efficiency, uncertainty, fitted value and residual. Untick **Use** for doubtful points and press **Refit**. Lines of cascade emitters (Ba-133, Y-88, Co-60, Eu-152 …) lose counts by summing in close geometries. Exclude them, or keep them and use the TCS correction of the activity calculation, which then corrects these calibration points as well (section 9.5).
5. Check χ^2/ν and the residual plot (points should scatter within their error bars), then **Save curve…** The curve (JSON) is stored in `%APPDATA%\SpectraMaster\efficiency\` and appears in the activity dialog.



	Nuclide	Activity (Bq)	Unc. %	Reference date	Half-life (s)	Lines keV:yield% (empty = library)
1	Am-241	671.074	3	1988-02-01 12:00	1.36641e+10	59.54:36.3
2	Cd-109	3654.35	4	1988-02-01 12:00	3.91621e+07	88.04:3.79
3	Co-57	148.364	2.5	1988-02-01 12:00	2.3328e+07	122.07:85.6
4	Ce-139	177.75	2.5	1988-02-01 12:00	1.188e+07	165.85:80
5	Hg-203	556.564	5	1988-02-01 12:00	4.02624e+06	279.17:81.5
6	Sn-113	717.002	5	1988-02-01 12:00	9.94464e+06	391.71:64.17
7	Sr-85	873.59	2.5	1988-02-01 12:00	5.59267e+06	513.99:99.28
8	Cs-137	619.83	3.5	1988-02-01 12:00	9.49863e+08	661.62:84.62
9	Y-88	1429.13	3.5	1988-02-01 12:00	9.21024e+06	898.02:94; 1836.01:99.36
10	Co-60	743.094	1	1988-02-01 12:00	1.66371e+08	1173.23:99.86; 1332.51:99.98

Figure 23 — Efficiency calibration of QCY48CAL.S0 from the Aptec source records: two-branch model, $\chi^2/\nu = 0.35$, ε(661.7 keV) = 0.00687 ± 2.6 %.

Point efficiency: $\varepsilon = N / (t_{live} \cdot A(t_0) \cdot C \cdot I)$, with the activity decay-corrected from the reference date to the start of counting and C the decay during counting. The curve uncertainty combines the fit covariance (scaled by χ^2/ν when > 1) with the part of the certificate uncertainty common to all lines.

9.2 Activity calculation

1. Open the sample spectrum (calibrated in energy) and choose **Analyze › Activity Calculation…**.
2. Select the **efficiency curve**. Optionally load a **background** spectrum: *peak by peak* (the background peak, scaled by live time, is subtracted from each sample peak) or *full spectrum*.
3. Enter the **sample quantity** and unit (kg, g, L, mL) with its uncertainty, and the **reference date** (sampling date; default = acquisition start). All activities are decay-corrected to this date.
4. List the **nuclides of interest**, choose *key line* or *weighted mean of lines*, the **detection limit** method (Currie or ISO 11929) and, if needed, the **self-attenuation correction factor** (default 1.000).
5. Optional corrections: a **reagent blank** (9.3), a **reference peak** (9.4) and **true-coincidence summing** (9.5). Fill in the **sample information** to get concentrations (9.6).
6. Click **Calculate**. The calculation runs in the background (the dialog shows *Calculating …*). **Export CSV/Excel…** saves the table and the uncertainty budget.

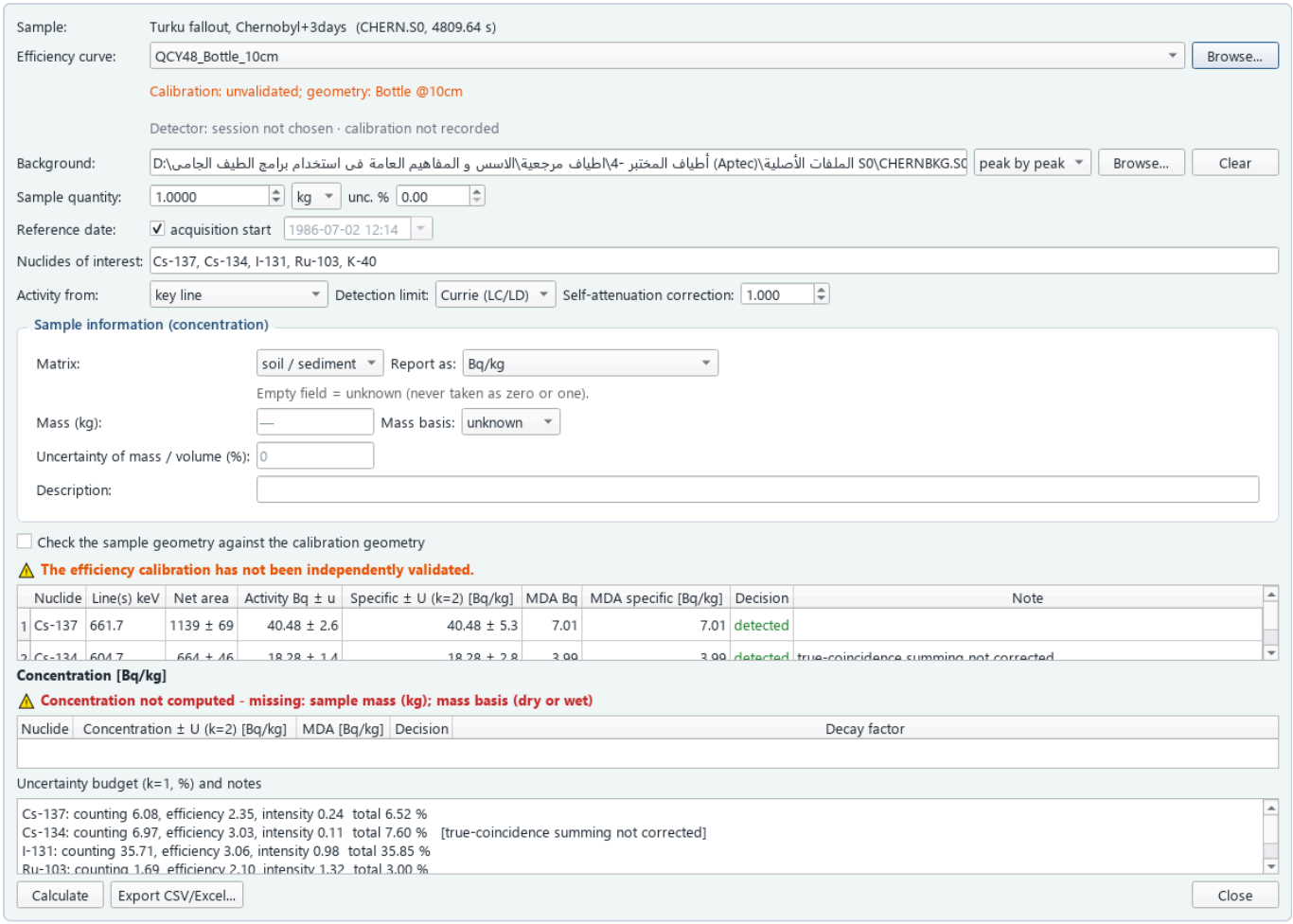


Figure 24 — Activities of CHERN.S0 (Turku fallout, 1986) with the QCY48 curve and peak-by-peak background subtraction.

Activity Bq ± u	Activity with its combined standard uncertainty (k = 1).
Specific ± U (k=2)	Activity per kg (or L …) with the **expanded** uncertainty U = 2u.
MDA	Minimum detectable activity (Bq and per unit quantity), from the continuum under the expected peak (Currie: $L_D = k^2 + 2L_C$, k = 1.645; ISO 11929: detection limit y#).
Decision	*detected* when the net signal exceeds the decision threshold (Currie L_C or ISO y*); otherwise < *MDA*. A nuclide can be *detected* with an activity below its MDA (between L_C and L_D) — report it with its uncertainty.
Budget	Relative components (k = 1): counting, efficiency, emission probability, quantity, attenuation, and, when used, the reagent blank, the reference peak and the summing correction.

If the sample spectrum was already background subtracted, stripped, normalized or smoothed, its channels are no longer Poisson counts. The results then carry a note that their uncertainties, decision thresholds and MDA are not valid. Analyse the measured spectrum with its background instead. A spectrum that was already subtracted is not subtracted again.

9.3 Reagent blank

A **reagent blank** is a blank sample prepared and counted like the samples, with the same reagents, container and geometry. Choose its spectrum in **Reagent blank**. It is analysed exactly like the sample, with the same curve, options and background, and its activity is subtracted line by line. The blank's uncertainty, including the part of its reference-peak correction it shares with the sample, is in the budget. A blank analysed with other options (attenuation, summing correction, reference peak, background) is reported in the notes. The session and the report record the blank file with its SHA-256 fingerprint.

9.4 Reference peak correction

The live time corrects dead time but not every loss: random summing and pile-up remove counts from every peak. A **reference peak**, from a pulser or a fixed reference source counted with every sample, measures this loss. Each result is multiplied by K = expected rate / measured rate of that peak in the sample spectrum.

- Enter the **peak energy** and its **expected rate** (counts per live second) with its uncertainty. **Measure from a spectrum…** takes the rate from a low-rate reference measurement, and its date becomes the rate date.
- For a radioactive reference source, give its **half-life** in days so the expected rate is decay corrected. Use 0 for a pulser.
- If the peak is not found, the calculation is refused. The reagent blank, when given, is corrected the same way.

9.5 True-coincidence summing correction

Nuclides that emit several gamma rays in cascade (Co-60, Cs-134, Bi-214, Y-88, Ba-133, Eu-152 …) lose counts from their peaks when two photons are detected together. Some lines also gain counts this way. The loss grows as the source gets closer to the detector. With **True-coincidence summing correction** enabled, each line of a cascade emitter is divided by its first-order coincidence summing factor (COI). The factor comes from gamma-gamma coincidence data and the efficiency curve.

- The total efficiency is the peak efficiency divided by the **peak-to-total ratio** PT(E) = min(1, a·(E / 1000 keV)^{b}). PT depends on the detector **and** the geometry. The values shown are assumptions; enter those of your detector, measured with single-line sources or taken from a detector model. Their uncertainty (default 10 %) is propagated.
- If the efficiency curve itself was measured with cascade emitters in the same geometry (for example a mixed standard with Ba-133 and Y-88), its points already contain summing losses. They are corrected with the same PT and the curve is refitted, so the summing is not counted twice. A curve in a model that cannot be refitted is refused.
- A line that the coincidence data do not describe is not corrected, and the note says why. A line whose first-order factor is outside its validity gives no result. A summing-out loss above 10 % is noted, because the first-order correction degrades in very close geometries.
- Not included: X-ray and 511 keV summing, and angular correlation. The session records the coincidence table with its SHA-256 fingerprint.

Command line: `--tcs-pt A,B` (and `--tcs-pt-unc`), `--blank FILE`, `--ref-peak keV --ref-rate cps` (or `--ref-spectrum FILE`, `--ref-half-life s`).

9.6 Sample information and concentrations

In **Sample information**, choose the **matrix** (soil / sediment, water, food, air filter, other) and the unit to report: Bq, Bq/kg, Bq/L or Bq/m³.

- **Bq/kg** needs the mass and its basis (dry or wet). **Bq/L** needs the volume.
- **Bq/m³** needs the air volume, either given directly or as flow rate × sampling duration. It also uses the measured fraction of the filter and the collection efficiency. The collection efficiency is an assumption, so state its source.
- An empty field means *unknown*: it is never taken as zero or one. A missing value is listed instead of a concentration.
- Non-detects are shown as < MDA, never as zero.

9.7 Worked example: QCY48CAL → CHERN (command line)

```
smconvert effcal QCY48CAL.S0 -o qcy48.json --model two_branch --compare
model: two-branch log-log, Ec = 583.2 keV, orders 3/2   chi2/dof = 0.35   u_sys = 1.00 %
nuclide       keV        net        eps    u %        fit  resid %
Am-241      59.54       3616  0.0050092   4.46  0.0049762     0.66
Cs-137     661.62      10909   0.007029   3.71   0.006877     2.21
Co-60     1332.51       6873    0.00315   2.03  0.0031219     0.90   ...
comparison with Aptec's stored efficiencies:  Cs-137 661.62  Aptec 0.0071482  ours 0.007029  -1.67 %

smconvert activity CHERN.S0 --eff qcy48.json --bkg CHERNBKG.S0 --nuclides Cs-137,Cs-134,I-131,Ru-103,K-
40 ^
          --mass 1 --unit kg -o chern.xlsx
Nuclide  Line(s) keV  Net area   Activity Bq (k=1)  Bq/kg (k=2)  MDA Bq  Decision
Cs-137   661.7        1157 ± 68  41.11 ± 2.6        41.11 ± 5.3  6.73    detected
Cs-134   604.7        683 ± 40   18.81 ± 1.3        18.81 ± 2.5  3.4     detected
I-131    364.5        120 ± 36   2.323 ± 0.71       2.323 ± 1.4  3.12    detected
Ru-103   497.1        5591 ± 90  138.5 ± 3.9        138.5 ± 7.8  3.99    detected
  Cs-134: true-coincidence summing not corrected
```

The last line appears because the summing correction was not requested. With `--tcs-pt A,B` the Cs-134 lines are corrected (section 9.5).

Without `--cert` the certificate is taken from the Aptec records in the `.S0` file; `--compare` lists the difference to Aptec's own stored efficiencies (here within ±6 %). A certificate CSV has the columns `nuclide,activity_bq,unc_percent,ref_date[,half_life_s,energy_kev,intensity_percent]`.

> **About this example.** The two files were measured in different set-ups (a 10 ml bottle on a 10 % HPGe and a rag wipe in another geometry); the example only shows the workflow. In practice the standard and the sample must share detector and geometry. Note also I-131: detected (above L_C) but below its MDA.

10. Printing and PDF reports

10.1 Report of the displayed spectrum

File › Print Report… (Ctrl+P), **Print Preview…** and **Export Report as PDF…** (Ctrl+Shift+P) first open the **Report Options** dialog. If the report should contain peaks or nuclides and the spectrum has not been analysed yet, the program offers to run the analysis.

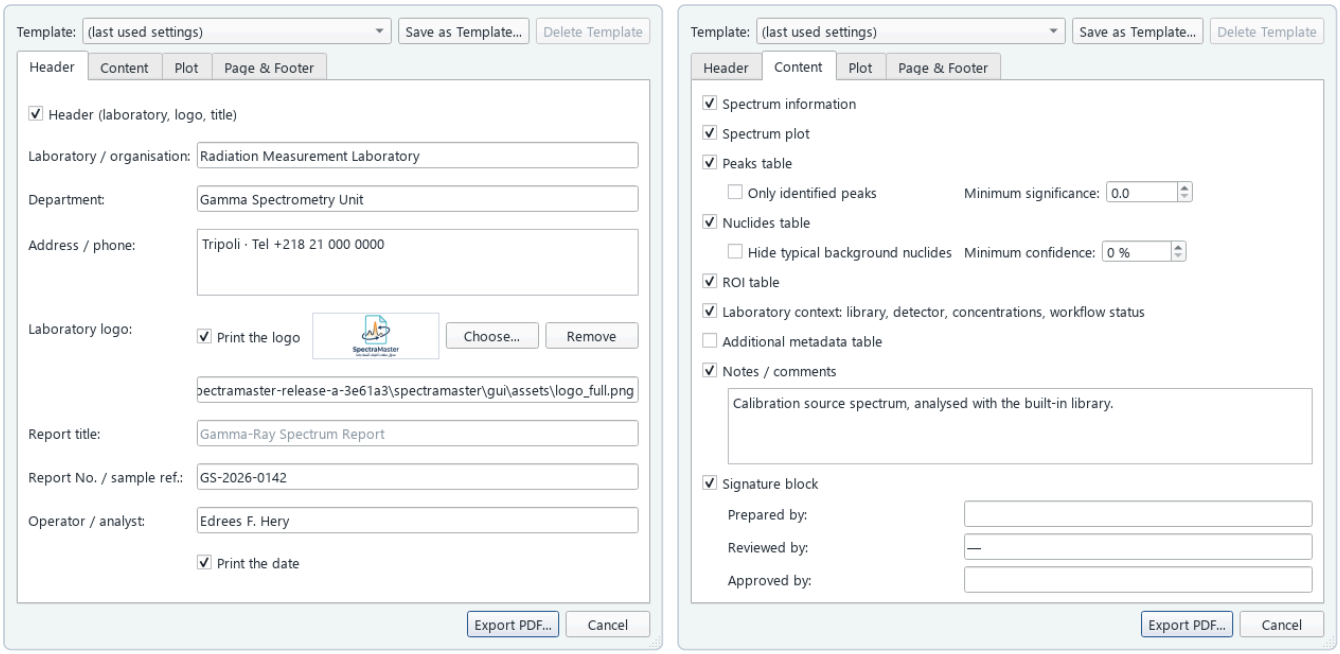
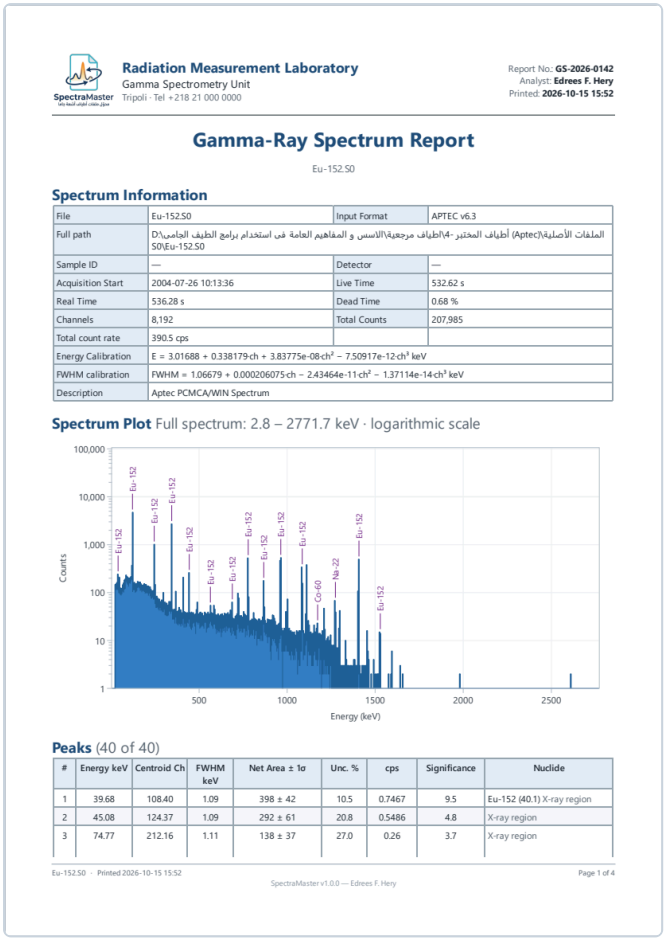


Figure 25 — Report options: Header and Content tabs.

Header	Laboratory, department, address/phone, laboratory logo, report title, report number / sample reference, analyst, date.
Content	Spectrum information, plot, peaks table (only identified peaks, minimum significance), nuclides table (minimum confidence, hide typical background nuclides), ROI table, additional metadata, notes, signature block (prepared / reviewed / approved).
Plot	Current zoom or full spectrum, counts scale, energy or channel axis, peak labels, ROI shading, overlays, plot height, vector graphics or 300 dpi image.
Page & Footer	Paper, orientation, margins, report language (Arabic, English or as the interface), footer items (page X of Y, file name, program credit, print date and time).



Radiation Measurement Laboratory
Gamma Spectrometry Unit
Tripoli · Tel +218 21 000 0000

Report No: **GS-2026-0142**
Analyst: **Edrees F. Hery**
Printed: **2026-10-15 15:52**

Gamma-Ray Spectrum Report

Eu-152.S0

Spectrum Information

File	Eu-152.S0	Input Format	APTEC v6.3
Full path	D:\أطياف مرجعية\الاسس و المفاهيم العامة فى استخدام برامج الطيف الجامى\4- أطياف المختبر (Aptec)\الملفات الأصلية\Eu-152.S0		
Sample ID	—	Detector	—
Acquisition Start	2004-07-26 10:13:36	Live Time	532.62 s
Real Time	536.28 s	Dead Time	0.68 %
Channels	8,192	Total Counts	207,985
Total count rate	390.5 cps		
Energy Calibration	E = 3.01688 + 0.338179·ch + 3.83775e-08·ch² − 7.50917e-12·ch³ keV		
FWHM calibration	FWHM = 1.06679 + 0.000206075·ch − 2.43464e-11·ch² − 1.37114e-14·ch³ keV		
Description	Aptec PCMCA/WIN Spectrum		

Spectrum Plot Full spectrum: 2.8 – 2771.7 keV · logarithmic scale

Peaks (40 of 40)

#	Energy keV	Centroid Ch	FWHM keV	Net Area ± 1σ	Unc. %	cps	Significance	Nuclide
1	39.68	108.40	1.09	398 ± 42	10.5	0.7467	9.5	Eu-152 (40.1) X-ray region
2	45.08	124.37	1.09	292 ± 61	20.8	0.5486	4.8	X-ray region
3	74.77	212.16	1.11	138 ± 37	27.0	0.26	3.7	X-ray region

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#	Energy keV	Centroid Ch	FWHM keV	Net Area ± 1σ	Unc. %	cps	Significance	Nuclide
4	121.74	351.06	1.25	16,615 ± 998	6.0	31.19	16.6	Eu-152 (121.8)
5	244.78	714.84	1.22	3,664 ± 99	2.7	6.879	36.9	Eu-152 (244.7)
6	248.01	724.41	1.22	258 ± 47	18.2	0.4844	5.5	
7	296.04	866.39	1.25	146 ± 30	20.6	0.2744	4.9	Eu-152 (295.9)
8	344.35	1009.23	1.26	10,600 ± 113	1.1	19.9	93.9	Eu-152 (344.3)
9	367.59	1077.95	1.29	273 ± 33	11.9	0.5121	8.4	Eu-152 (367.8)
10	411.10	1206.60	1.32	700 ± 32	4.6	1.314	21.7	Eu-152 (411.1)
11	443.95	1303.70	1.34	947 ± 37	3.9	1.778	25.4	Eu-152 (444.0)
12	488.83	1436.38	1.36	118 ± 20	17.0	0.2216	5.9	Eu-152 (488.7)
13	564.10	1658.91	1.41	108 ± 21	19.6	0.202	5.1	Eu-152 (564.0)
14	586.32	1724.62	1.42	96 ± 19	20.0	0.1804	5.0	Eu-152 (586.3)
15	678.75	1997.89	1.48	96 ± 21	21.5	0.1805	4.7	Eu-152 (678.6)
16	688.63	2027.09	1.48	187 ± 22	11.5	0.3515	8.7	Eu-152 (688.7)
17	723.33	2129.68	1.51	316 ± 25	7.8	0.594	12.8	Zr-95 (724.2)
18	756.78	2228.58	1.53	64 ± 19	29.1	0.1205	3.4	Zr-95 (756.7)
19	778.86	2293.87	1.60	2,609 ± 57	2.2	4.898	45.9	Eu-152 (778.9)
20	810.42	2387.17	1.56	66 ± 17	25.9	0.1244	3.9	Eu-152 (810.5)
21	867.36	2555.50	1.59	784 ± 33	4.2	1.471	23.5	Eu-152 (867.4)
22	873.12	2572.53	1.60	129 ± 19	15.0	0.2421	6.7	
23	919.39	2709.33	1.62	75 ± 16	20.8	0.1416	4.8	Eu-152 (919.3)
24	926.01	2728.91	1.63	70 ± 19	26.8	0.1315	3.7	Eu-152 (926.3)
25	964.10	2841.54	1.66	2,615 ± 55	2.1	4.909	47.3	Eu-152 (964.1)
26	996.37	2936.94	1.67	111 ± 17	15.3	0.2075	6.5	
27	1004.99	2962.44	1.68	323 ± 22	6.9	0.6072	14.5	Eu-152 (1005.3)
28	1085.79	3201.33	1.72	1,734 ± 54	3.1	3.256	32.2	Eu-152 (1085.8)
29	1089.65	3212.77	1.72	304 ± 32	10.4	0.5707	9.6	Eu-152 (1089.7)
30	1112.07	3279.05	1.81	2,170 ± 68	3.2	4.074	31.7	Eu-152 (1112.1)
31	1123.07	3311.57	1.75	77 ± 14	18.7	0.1445	5.3	sum peak (344+779)
32	1173.33	3460.19	1.78	46 ± 15	31.8	0.08604	3.1	Co-60 (1173.2)
33	1212.99	3577.48	1.80	166 ± 18	10.6	0.312	9.5	Eu-152 (1212.9)
34	1274.36	3758.95	1.84	330 ± 23	7.0	0.6191	14.2	Na-22 (1274.5)
35	1299.18	3832.36	1.86	222 ± 18	8.3	0.4165	12.1	Eu-152 (1299.1)
36	1332.73	3931.57	1.88	35 ± 8	22.7	0.06579	4.4	Co-60 (1332.5)
37	1408.06	4154.36	2.02	2,875 ± 59	2.0	5.398	48.8	Eu-152 (1408.0)
38	1457.59	4300.85	1.95	56 ± 11	19.1	0.1052	5.2	Eu-152 (1457.6)
39	1529.16	4512.57	2.00	93 ± 13	14.1	0.1753	7.1	Eu-152 (1528.1) sum peak (122+1408)
40	1596.62	4712.12	2.04	22 ± 5	23.5	0.04119	4.3	Ba-140 (1596.2)

Identified Nuclides (6 of 6) · Library: Environmental

Nuclide	Confidence	Half-life	Category	Lines Found / Expected	Lines found (keV)	Note
Eu-152	94 %	13.517 y	Activation	28 / 28	40.1, 121.8, 244.7, 295.9,	

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Figure 26 — First two pages of a full report (Eu-152.S0).

10.2 Templates

The dialog always starts with the last used settings. **Save as Template…** stores the current options under a name (e.g. "Food samples – Arabic"); choose it later in the **Template** list. Built-in presets: *Full report*, *Plot only*, *Nuclides summary*. **Delete Template** removes a user template.

10.3 Reports for many files

File › Print Reports for Multiple Files…: add files or a whole folder, set the **Report Options…** once, choose **One PDF per spectrum file** (into an output folder) or **One combined PDF**, and optionally **Run peak search & identification for every file**. Click **Create PDFs**; the list shows the pages written or the error for each file.



Figure 27 — Batch reports.

11. Archive search

File › Archive Search… (Ctrl + Shift + F) finds spectra in large archives — by sample name, words in the description, nuclides found, date, detector, folder or live time — without moving or changing the original files.

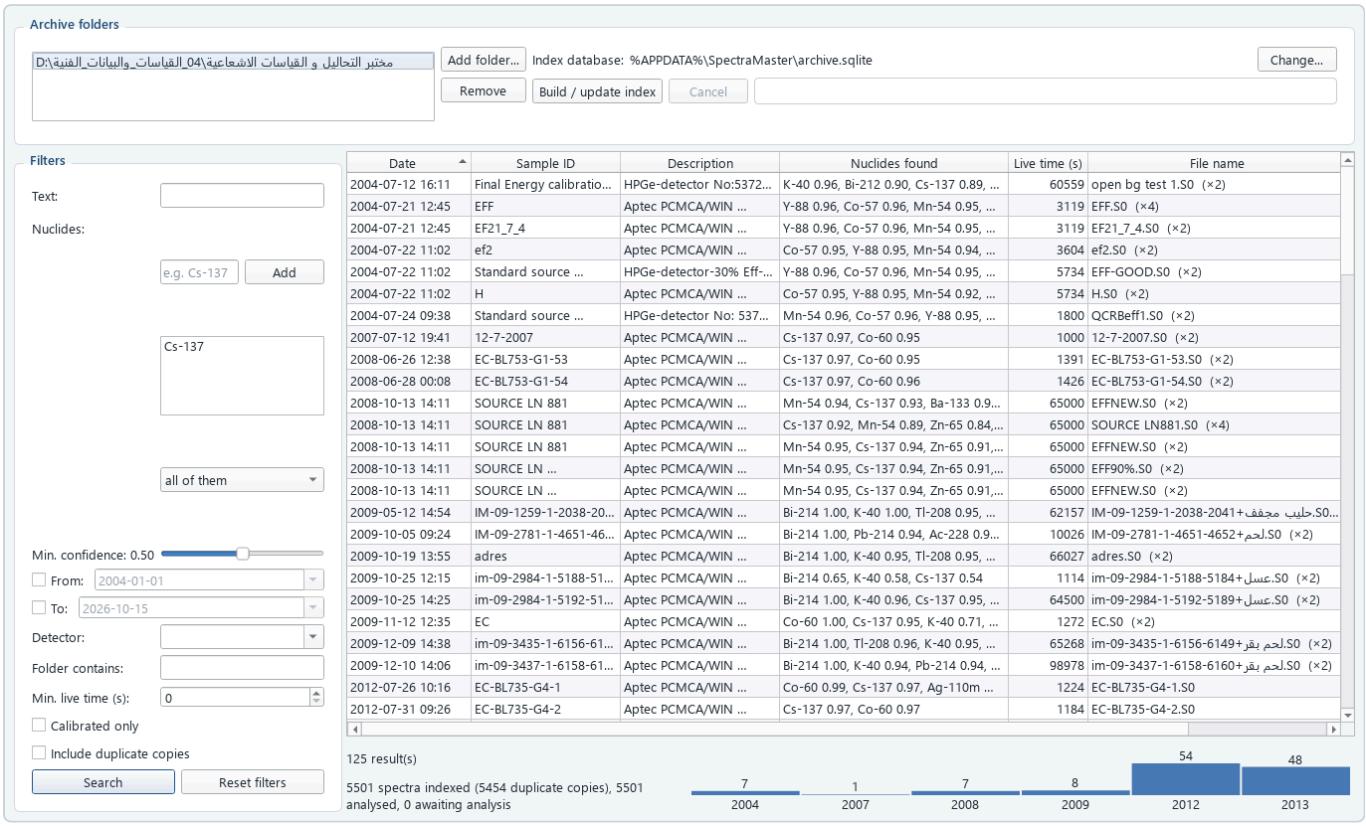


Figure 28 — Archive search on an index of 5,501 unique spectra (5,454 duplicate copies): all spectra with Cs-137 (confidence ≥ 0.50), results per year below.

11.1 Building and updating the index

1. **Add folder…** for each archive root (local disk or network share).
2. Click **Build / update index**. The program lists the files, reads the metadata of every spectrum and then analyses each unique spectrum (peak search + identification) in parallel. Progress and the remaining time are shown; **Cancel** stops safely and the next build resumes.
3. Later builds only process new or changed files and drop deleted ones, so updating is quick.

The index is a single SQLite file, by default `%APPDATA%\SpectraMaster\archive.sqlite` (**Change…** selects another file, e.g. a shared copy on the network). Byte-identical copies of a spectrum are recognised as duplicates and analysed once. The original files are only read.

11.2 Filters

Text	Words in the sample ID, description, comments, file or folder name. All words must match.
Nuclides	Add one or more nuclides (with auto-completion: `Cs-137`, `cs137`, `137Cs` are equivalent); match *all of them* or *any of them*, with a **minimum confidence** slider.
Date, detector, folder	From / to date, detector name, part of the folder path.
Other	Minimum live time, calibrated spectra only, include duplicate copies.

> **Arabic normalisation.** The text search ignores the differences أ إ آ → ا, ة → ه, ى → ي, diacritics (tashkeel) and tatweel, and treats Arabic-Indic digits like Latin digits — `شاى` finds `شاي`. The definite article is removed from the search words (also وال، بال، كال، فال، لل) when at least three letters remain, so `الشاي` finds `شاي`, `الشاي` and `بالشاي` alike.

11.3 Working with the results

Double-click a row to open the spectrum in the main window. The right-click menu offers **Open in main window**, **Add as overlay**, **Open containing folder** and **Copy path**. Columns can be sorted; the bar chart shows the results per year. **Export results to Excel / CSV…** saves the list with date, sample, description, nuclides, times, detector, path and number of copies.

The same index can be built from the command line for very large archives with `smconvert index` (Excel catalogue) — the window's index is built from the window itself.

12. Detector quality control

Tools › Detector QC… keeps control charts of the check-source and background measurements of each detector, as required by ISO/IEC 17025 (7.7). For every measurement it records the **centroid energy** (gain/offset drift), the **FWHM** (resolution), the **net count rate** decay-corrected to a reference date (efficiency stability), the **background rate** in an energy window, and live/dead time.

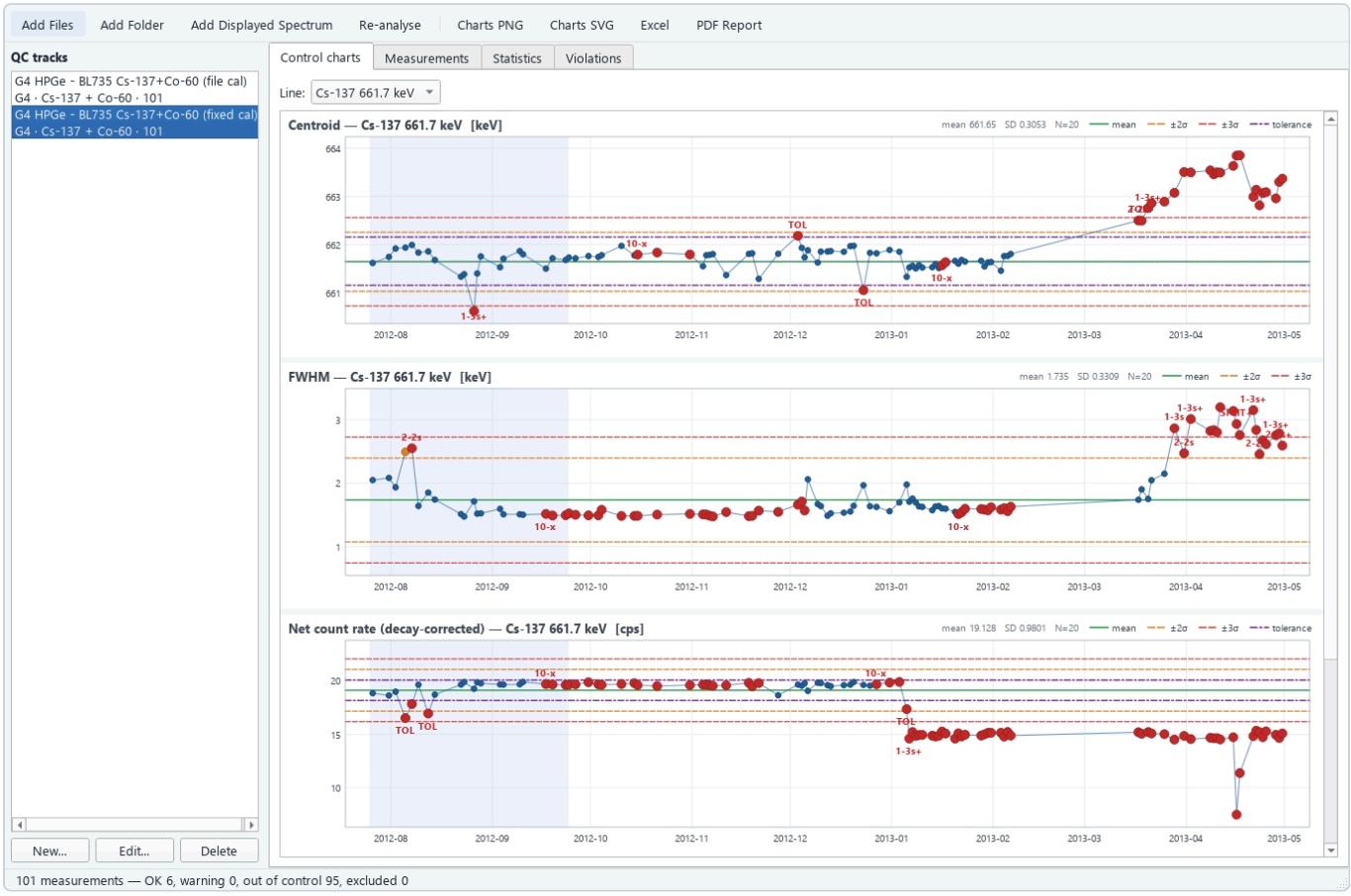


Figure 29 — Control charts of a Cs-137 + Co-60 check source on detector G4 (101 measurements, 2012–2013): baseline (shaded), mean, ±2σ, ±3σ and tolerance lines, violations labelled.

12.1 Creating a QC track

A **track** is one detector + one check source (or a background measurement). Click **New…** and fill in:

- name, detector, detector type (HPGe, NaI, LaBr3, CZT), type (check source / background);
- **monitored lines** — from a preset (Cs-137, Co-60, Cs-137 + Co-60, Eu-152, Ba-133, Am-241, Na-22, Background) or typed as `keV[:nuclide[:FWHM max]]`, e.g. `661.657:Cs-137, 1332.492:Co-60:2.0`;
- reference date and reference activity (for decay correction), peak match tolerance;
- **centroid energy from** each file's calibration, or a **fixed reference calibration** — the latter shows real gain drift even when each spectrum was recalibrated;
- background window, **baseline** (first N points or a period), the **Westgard rules** to apply and fixed tolerances (centroid ± keV, rate ± %, background ± %).

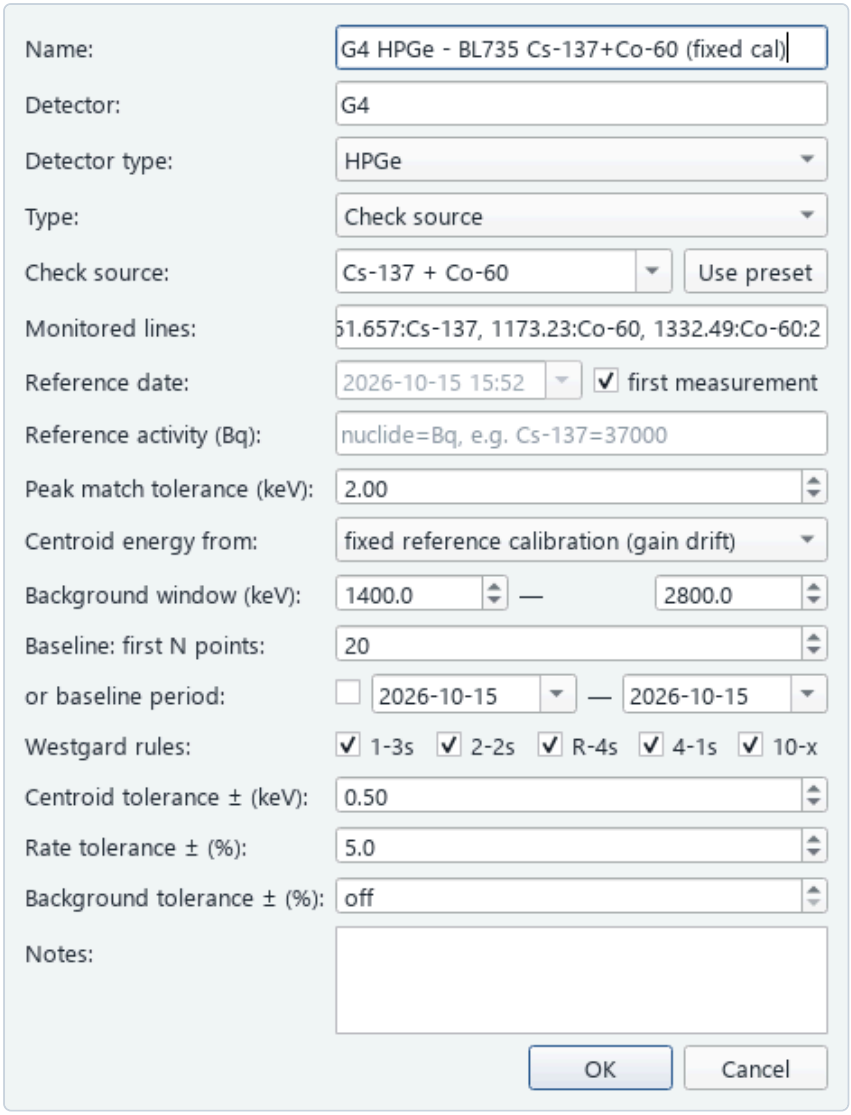


Figure 30 — QC track settings.

12.2 Adding measurements

Add Files…, **Add Folder…** (e.g. the whole year's QC folder) or **Add Displayed Spectrum**. Each spectrum is analysed once and its peak list stored, so lines and tolerances can be changed later without re-reading the files; duplicates are ignored. **Re-analyse** re-reads all source files. In the *Measurements* table a right click can **exclude** a point from the statistics (with a note, e.g. "source repositioned"), remove it, or show the spectrum in the main window.

12.3 Control charts and Westgard rules

Mean and standard deviation are computed from the baseline points; warning limits are ±2σ and action limits ±3σ. The rules are evaluated on all non-excluded points in date order:

Rule	Meaning	Severity
`1-2s`	one point beyond ±2σ	warning
`1-3s`	one point beyond ±3σ	out of control
`2-2s`	two consecutive points beyond 2σ on the same side	out of control

Rule	Meaning	Severity
`R-4s`	two consecutive points beyond 2σ on opposite sides	out of control
`4-1s`	four consecutive points beyond 1σ on the same side	out of control
`10-x`	ten consecutive points on the same side of the mean	out of control
`TOL`	fixed tolerance exceeded (centroid, FWHM max, rate, background)	out of control
`SPLIT`	line found split into two peaks (gain or zero jump during the count)	out of control

The *Violations* tab lists every rule hit with date, file, series and value; the *Statistics* tab gives N, mean, SD, CV % and the trend per year for each line and quantity.

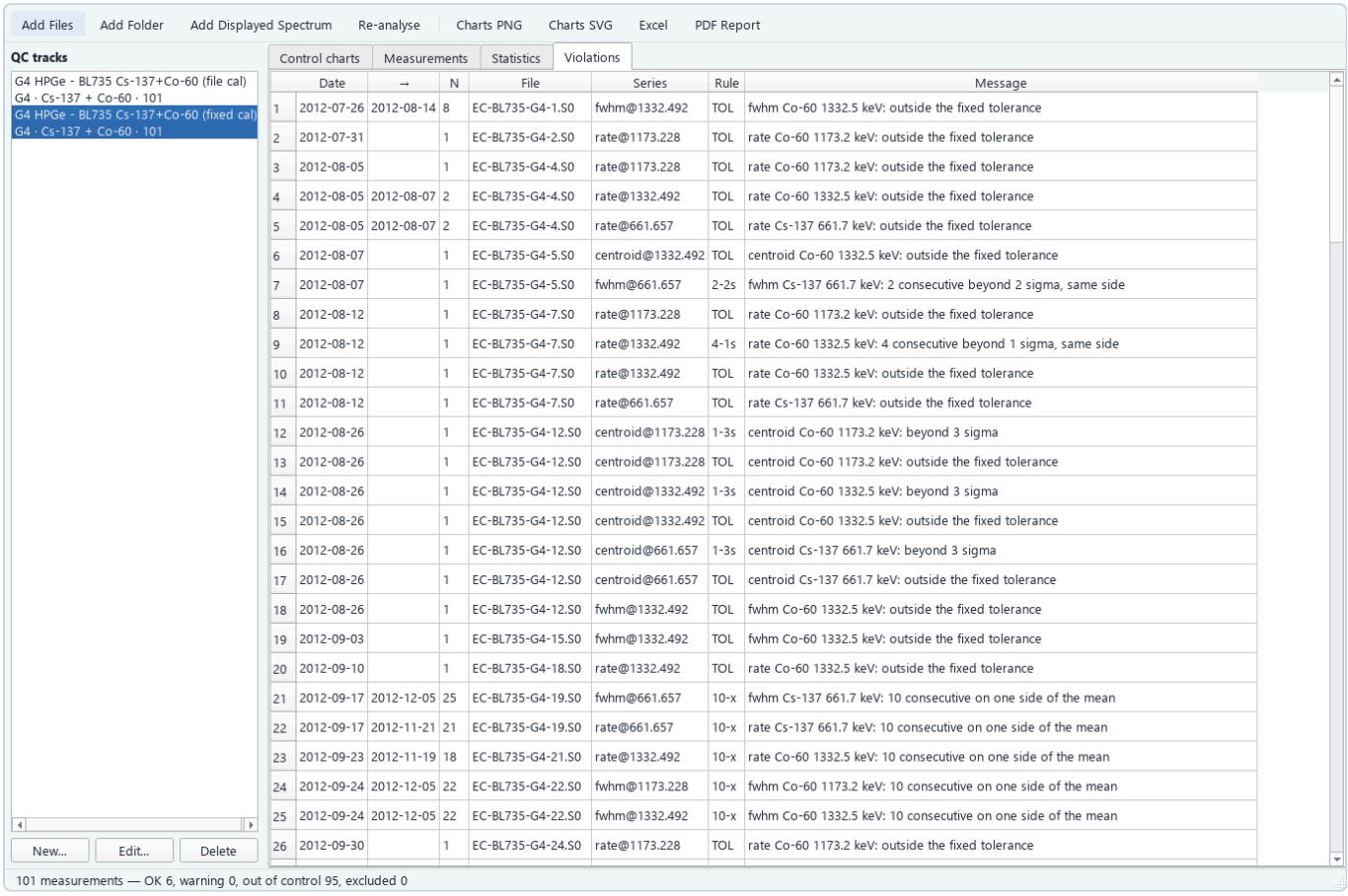


Figure 31 — Violations tab.

12.4 Exporting and reports

Charts PNG… / SVG… save the charts, **Excel…** a workbook with measurements, statistics and violations, and **PDF Report…** a complete QC report (track settings, summary, charts, statistics, measurements, violations and a signature line) in the interface language. Tracks are stored as JSON files in `%APPDATA%\SpectraMaster\qc\`.

13. Troubleshooting and FAQ

The file is not recognised ("cannot recognise the format")

Check that the file is a spectrum (not a report or a calibration file) and not empty. Try **File › Open Input Files As...** with the expected format; the `smconvert info FILE --from CODE` command shows the reader's exact error. If *Auto-detect* is not active, a wrong fixed format is the most common cause — the status bar shows the active input mode.

Peaks are at the wrong energies

- The file may be uncalibrated or carry an old calibration: check *Energy Calibration* in the information panel and recalibrate (chapter 8).
- After conversion to CHN/CNF/SPC, read the refit warning: the deviation of the refitted calibration is printed.
- Genie-2000 shows channel numbers one higher than SpectraMaster — energies are the same.

Arabic sample names became "????" in Genie or MAESTRO

CNF, CHN and SPC store only Latin text (section 3.4). Keep the Aptec S0, SPE, N42 or Excel file as the reference copy. Saving or converting to a format that stores only ASCII text (CHN, SPC, CNF, S100, ND66) prints a warning when the sample ID or description contains Arabic (non-ASCII) characters.

"stores only 3 calibration terms: energy calibration refitted"

Not an error: the target format cannot hold the source polynomial. Accept small deviations ($\lesssim 0.05$ keV); for larger ones choose a format that keeps the full calibration (Aptec S0, IAEA SPE, N42, JSON, Excel).

"The peak-search engine is not available" / P is disabled

No spectrum is open, the spectrum is empty, or (SpectraMaster method) the analysis engine is missing. Nuclides are not identified in uncalibrated spectra.

GammaVision does not call SpectraMaster

Copy Program and Arguments again from **Help › GammaVision / MAESTRO Integration...**; put quotes around paths with spaces when not using the macros.

Where are settings and data stored?

Preferences	Language, last folders, input/output formats, window layout, view options and report templates: Windows registry `HKEY_CURRENT_USER\Software\SpectraMaster\SpectraMaster`.
Archive index	`%APPDATA%\SpectraMaster\archive.sqlite` (changeable in the window).
Efficiency curves	`%APPDATA%\SpectraMaster\efficiency\*.json`
QC tracks	`%APPDATA%\SpectraMaster\qc\*.json`

`%APPDATA%` is usually `C:\Users\<name>\AppData\Roaming`. Back up the `SpectraMaster` folder together with the laboratory data. Deleting the registry key resets all preferences.

14. Appendices

A. Supported formats

Codes 1–16 are the DataMaster numbers; use them with `-i` / `-j`, `--to` and `--from`.

60 formats — 59 readable, 35 writable

Code	Key	Format	Extensions	Read	Write
1	`ortec_chn`	ORTEC ADCAM / MAESTRO	*.chn	✓	✓
2	`ortec_spc`	ORTEC INFORM (integer SPC)	*.spc	✓	✓
5	`ascii1`	ASCII-1 (one count per line)	*.txt	✓	✓
6	`ascii2`	ASCII-2 (channel, counts)	*.txt	✓	✓
7	`iaea_spe`	IAEA / Rossendorf ASCII SPE	*.spe	✓	✓
8	`iec1455`	IEC 1455 ASCII	*.iec	✓	✓
9	`ieee_asc`	IEEE ASCII	*.asc	✓	✓
10	`rms_phd`	RMS / IMS ASCII PHD	*.phd	✓	—
13	`aptec_s0`	APTEC v6.3	*.s0	✓	✓
14	`ci_s100`	Canberra S100	*.mca	✓	✓
15	`ci_accuspec`	Canberra AccuSpec	*.dat	✓	—
16	`ci_cnf`	Canberra CAM (Genie-2000)	*.cnf	✓	✓
17	`eml_sanderson`	EML/Sanderson ASCII test spectra	*.dat	✓	—
20	`n42_2006`	ANSI N42.42-2006 XML	*.n42 *.xml	✓	✓
21	`n42_2012`	ANSI N42.42-2012 XML	*.n42 *.xml	✓	✓
22	`gadras_pcf`	GADRAS PCF	*.pcf	✓	✓
23	`specutils_csv`	SpecUtils CSV (Energy, Data)	*.csv	✓	✓
24	`specutils_txt`	SpecUtils TXT	*.txt	✓	✓
25	`tka`	Canberra/Toolkit TKA	*.tka	✓	✓
26	`html_d3`	HTML page (interactive D3 chart)	*.html *.htm	—	✓
27	`spc_ascii`	ORTEC/IAEA ASCII SPC	*.spc	✓	✓
28	`spc_float`	ORTEC SPC (floating point)	*.spc	✓	✓
29	`exploranium_gr130`	Exploranium GR-130 binary	*.dat	✓	✓
30	`exploranium_gr135`	Exploranium GR-135 v2 binary	*.dat	✓	✓
31	`amptek_mca`	Amptek MCA (read only)	*.mca	✓	—
32	`radiacode_xml`	RadiaCode / BecqMoni XML (read only)	*.xml *.rcspg	✓	—
33	`lsrm_spe`	LSRM / SpectraLine SPE (read only)	*.spe	✓	—
34	`multiact`	MultiAct (read only)	*.spm	✓	—
35	`labzy_lzs`	LabZY nanoMCA LZS (read only)	*.lzs	✓	—
36	`caen_gxml`	CAEN Hexagon GXML (read only)	*.gxml	✓	—
37	`specutils_json`	SpecUtils JSON (read only)	*.json	✓	—
38	`tracs_mps`	TRACS MPS (read only)	*.mps	✓	—
39	`aram`	ARAM (read only)	*.xml	✓	—
40	`microraider`	ICx MicroRaider XML (read only)	*.xml	✓	—
41	`spm_daily`	Spectroscopic portal daily file (read only)	*.txt	✓	—
42	`scandata_xml`	ScanData XML (read only)	*.xml	✓	—

Code	Key	Format	Extensions	Read	Write
43	`ortec_listmode`	ORTEC list mode (read only, summed)	*.lis	✓	—
50	`excel_xlsx`	Microsoft Excel workbook	*.xlsx	✓	✓
51	`json`	SpectraMaster JSON	*.json	✓	✓
60	`aptec_ascii`	Aptec ASCII	*.apt	✓	✓
61	`bti_ascii`	BTI ASCII (read only)	*.bti	✓	—
62	`chandata_csv`	Aptec Channel Data CSV	*.csv	✓	✓
63	`ci90_ascii`	Canberra Series 90 ASCII (read only)	*.o	✓	—
64	`davidson_ascii`	Davidson ASCII	*.dav	✓	✓
65	`iaea_ascii_dat`	IAEA ASCII (Aptec, EML layout)	*.dat	✓	✓
66	`npl_ascii`	NPL ASCII	*.txt	✓	✓
67	`rump_ascii`	Rump ASCII (RBS)	*.mac	✓	✓
68	`wallac_ascii`	Wallac ASCII (1024 ch)	*.001	✓	✓
69	`laval_ascii`	Laval ASCII	*.de1	✓	✓
70	`jcac_vax`	JCAC VAX	*.spc	✓	✓
71	`nd66`	Nuclear Data ND66	*.001	✓	✓
72	`norland5000`	Norland PHA-5000	*.dat	✓	✓
73	`oxford_ans`	Oxford ANS	*.ans	✓	✓
74	`quantrad_scout`	Quantrad Scout (read only)	*.sct	✓	—
75	`spectran_f`	Spectran F (read only)	*.spc	✓	—
76	`nucleus_pca`	Nucleus PCA-I/PCA-II (read only)	*.spm	✓	—
77	`pca_multiport`	PCA-Multiport (read only)	*.spt	✓	—
78	`aptec_v30`	APTEC v3.0 (read only)	*.s0	✓	—
79	`aptec_v43`	APTEC v4.3 (read only)	*.s0	✓	—
80	`lab_dat1670`	EMCA2000 binary .dat (read only)	*.dat	✓	—

B. Command-line reference

Output of `smconvert --help` and `smconvert COMMAND --help`. Format arguments accept a code, key or alias. `smconvert` without arguments prints the usage summary and a reminder that the graphical program is `SpectraMaster.exe`.

```
usage: smconvert [-h] COMMAND ...

Gamma-spectrum file converter (DataMaster compatible).

positional arguments:
  COMMAND
    convert    convert files, folders or glob patterns
    info       print the metadata of spectrum files
    formats    list supported formats and their codes
    peaks      search and fit peaks, print a peak table
    identify   identify nuclides (peak search + library matching)
    gui        start the graphical interface
    index      catalogue a spectrum archive into an Excel workbook
    rescue     convert + verify a whole archive (resumable)
    sum        sum spectra (energy-aligned if calibrations differ)
    subtract   subtract a live-time normalised background
    rebin      compress channels (x2, x4 ...) and/or truncate the range
    calibrate  fit / propose / apply an energy (and FWHM) calibration
    batch      group spectra by detector/channels/calibration; analyse one
               group
    effcal     efficiency calibration from a certified standard spectrum
    activity   activity, specific activity and MDA of a sample spectrum

options:
  -h, --help   show this help message and exit

DataMaster form: spectramaster -i INPUT -j OUTPUT (i, j = format codes from
'spectramaster formats'; i = 0 auto-detects)
```

convert

```
usage: smconvert convert [-h] --to TO [--from IN_FMT] [--out OUT]
                         [--recursive] [--keep-tree] [--no-overwrite]
                         [--quiet]
                         inputs [inputs ...]

positional arguments:
  inputs             input files, folders or glob patterns

options:
  -h, --help         show this help message and exit
  --to, -t TO        output format key, alias or code
  --from, -f IN_FMT  input format key/code, or 'auto' (default)
  --out, -o OUT      output folder (default: next to each input)
  --recursive, -r    walk sub-folders
  --keep-tree        recreate the input sub-folder structure inside --out
  --no-overwrite     skip existing outputs (reported as failures)
  --quiet, -q        only print failures and the summary
```

info

```
usage: smconvert info [-h] [--from IN_FMT] files [files ...]

positional arguments:
  files

options:
  -h, --help         show this help message and exit
  --from, -f IN_FMT
```

formats

```
usage: smconvert formats [-h]

options:
  -h, --help  show this help message and exit
```

peaks

```
usage: smconvert peaks [-h] [--from IN_FMT] [--min-sig MIN_SIG]
                       files [files ...]

positional arguments:
  files

options:
  -h, --help         show this help message and exit
  --from, -f IN_FMT
  --min-sig MIN_SIG  significance threshold (default 3)
```

identify

```
usage: smconvert identify [-h] [--from IN_FMT] [--min-sig MIN_SIG]
                          [--lib LIB] [--min-conf MIN_CONF] [--peaks]
                          files [files ...]

positional arguments:
  files

options:
  -h, --help           show this help message and exit
  --from, -f IN_FMT
  --min-sig MIN_SIG    peak significance threshold (default 3)
  --lib LIB            nuclide library (JSON like the bundled one, or CSV
                       nuclide,energy_keV,intensity_%[,half_life,category])
  --min-conf MIN_CONF  minimum confidence 0..1 (default 0.25)
  --peaks              also print the peak table with assignments
```

gui

```
usage: smconvert gui [-h] [files ...]

positional arguments:
  files

options:
  -h, --help  show this help message and exit
```

index

```
usage: smconvert index [-h] -o OUT [--no-recursive] [--all-files]
                       [--quiet]
                       root

positional arguments:
  root            archive folder (read only)

options:
  -h, --help      show this help message and exit
  -o, --out OUT   catalogue .xlsx (outside the archive)
  --no-recursive  do not walk sub-folders
  --all-files     sniff every file, not only spectrum extensions
  --quiet, -q
```

rescue

```
usage: smconvert rescue [-h] -o OUT [--to TO] [--resume] [--flat]
                        [--no-verify] [--no-recursive] [--skip-duplicates]
                        [--limit LIMIT] [--quiet]
                        root

positional arguments:
  root               archive folder (read only)

options:
  -h, --help         show this help message and exit
  -o, --out OUT      output folder (outside the archive)
  --to, -t TO        output format (repeatable; default n42_2012 + ci_cnf)
  --resume           skip files already converted and verified
  --flat             do not recreate the folder tree
  --no-verify        skip read-back verification
  --no-recursive
  --skip-duplicates  convert byte-identical copies only once
  --limit LIMIT      only the first N candidate files (trial run)
  --quiet, -q
```

sum

```
usage: smconvert sum [-h] -o OUT [--to TO] [--align {auto,channel,energy}]
                     files [files ...]

positional arguments:
  files

options:
  -h, --help            show this help message and exit
  -o, --out OUT
  --to, -t TO           output format (default: from the extension / first
                        input)
  --align {auto,channel,energy}
```

subtract

```
usage: smconvert subtract [-h] -o OUT [--to TO] [--scale SCALE] [--clip]
                          [--sigma-csv SIGMA_CSV]
                          sample background

positional arguments:
  sample
  background

options:
  -h, --help            show this help message and exit
  -o, --out OUT
  --to, -t TO
  --scale SCALE         background factor (default LT_sample / LT_background)
  --clip                set negative net channels to 0
  --sigma-csv SIGMA_CSV
                        also write channel, energy, net, sigma to this CSV
```

rebin

```
usage: smconvert rebin [-h] -o OUT [--factor FACTOR] [--start START]
                       [--stop STOP] [--renumber] [--to TO]
                       file

positional arguments:
  file

options:
  -h, --help       show this help message and exit
  -o, --out OUT
  --factor FACTOR
  --start START    first channel to keep (truncate)
  --stop STOP      channel after the last one kept
  --renumber       renumber truncated channels from 0
  --to, -t TO
```

calibrate

```
usage: smconvert calibrate [-h] [--pairs PAIRS] [--source SOURCE]
                           [--coeffs COEFFS] [--order ORDER] [--fwhm FWHM]
                           [--fwhm-order FWHM_ORDER] [-o OUT] [--to TO]
                           [--only-uncalibrated]
                           files [files ...]

positional arguments:
  files

options:
  -h, --help            show this help message and exit
  --pairs PAIRS         channel:keV pairs, e.g. 1893.7:661.66,3364:1173.23
  --source SOURCE       calibration source nuclide(s), e.g. Eu-152 or
                        Co-60,Cs-137
  --coeffs COEFFS       apply these coefficients c0,c1[,c2,c3]
  --order ORDER         polynomial order 1-3
  --fwhm FWHM           channel:FWHM_keV pairs for a FWHM calibration
  --fwhm-order FWHM_ORDER
  -o, --out OUT         write recalibrated copies into this folder
  --to, -t TO           output format (default: input format if writable, else
                        IAEA SPE)
  --only-uncalibrated   leave already calibrated files alone
```

batch

```
usage: smconvert batch [-h] [--group GROUP] [--eff EFF]
                       [--geometry GEOMETRY] [--allow-geometry-mismatch]
                       [--application APPLICATION] [--nuclides NUCLIDES]
                       [--energy-tol ENERGY_TOL] [-o OUT]
                       paths [paths ...]

positional arguments:
  paths                 files, folders or glob patterns

options:
  -h, --help            show this help message and exit
  --group GROUP         group to analyse (see the list printed first)
  --eff EFF             efficiency curve JSON for this group (omit:
                        identification only)
  --geometry GEOMETRY   sample geometry JSON
                        (lab.detector.MeasurementGeometry)
  --allow-geometry-mismatch
                        use a calibration of another geometry anyway (reported
                        in every row)
  --application APPLICATION
                        identification library: all, environmental,
                        calibration, safeguards, medical
  --nuclides NUCLIDES   comma-separated nuclides for activities (default:
                        those identified)
  --energy-tol ENERGY_TOL
                        calibration agreement for grouping, keV
  -o, --out OUT
```

`effcal`

```
usage: smconvert effcal [-h] [--cert CERT] -o OUT
                        [--model {loglog,two_branch}] [--order ORDER]
                        [--order-high ORDER_HIGH] [--crossover CROSSOVER]
                        [--u-sys U_SYS] [--bkg BKG]
                        [--min-intensity MIN_INTENSITY]
                        [--geometry GEOMETRY] [--detector DETECTOR]
                        [--name NAME] [--compare]
                        file

positional arguments:
  file                  calibration (standard) spectrum

options:
  -h, --help            show this help message and exit
  --cert CERT           certificate CSV: nuclide,activity_bq,unc_percent,ref_d
                        ate[,half_life_s,energy_kev,intensity_percent]
                        (default: Aptec records in an .S0 file)
  -o, --out OUT         efficiency curve JSON
  --model {loglog,two_branch}
  --order ORDER         log-log order, or low-branch order (default 3)
  --order-high ORDER_HIGH
                        high-branch order (two_branch)
  --crossover CROSSOVER
                        crossover keV (two_branch; default: automatic)
  --u-sys U_SYS         correlated relative uncertainty (fraction)
  --bkg BKG             background spectrum (peak-by-peak subtraction)
  --min-intensity MIN_INTENSITY
                        library lines >= this % (default 2)
  --geometry GEOMETRY
  --detector DETECTOR
  --name NAME
  --compare             compare with Aptec's stored efficiencies
```

`activity`

```
usage: smconvert activity [-h] --eff EFF [--bkg BKG]
                          [--bkg-mode {peak,spectrum}]
                          [--ref-peak REF_PEAK] [--ref-rate REF_RATE]
                          [--ref-rate-unc REF_RATE_UNC]
                          [--ref-spectrum REF_SPECTRUM]
                          [--ref-half-life REF_HALF_LIFE] [--tcs-pt A,B]
                          [--tcs-pt-unc TCS_PT_UNC] [--blank BLANK]
                          [--mass MASS] [--unit UNIT]
                          [--mass-unc MASS_UNC] [--ref-date REF_DATE]
                          [--nuclides NUCLIDES] [--method {key,mean}]
                          [--mda {currie,iso11929}]
                          [--attenuation ATTENUATION] [-o OUT]
                          file

positional arguments:
  file

options:
  -h, --help            show this help message and exit
  --eff EFF             efficiency curve JSON (from 'effcal')
  --bkg BKG             background spectrum
  --bkg-mode {peak,spectrum}
  --ref-peak REF_PEAK   reference peak correction: energy keV of a pulser or
                        fixed reference source peak
  --ref-rate REF_RATE   its expected rate, counts per live second
  --ref-rate-unc REF_RATE_UNC
                        its relative uncertainty %
  --ref-spectrum REF_SPECTRUM
                        reference (low-rate) measurement giving the expected
                        rate
  --ref-half-life REF_HALF_LIFE
                        half-life (s) of a reference source (decay correction)
  --tcs-pt A,B          true-coincidence summing correction with the peak-to-
                        total ratio PT(E) = min(1, A (E/1000 keV)^B) of this
                        detector and geometry (an assumption)
  --tcs-pt-unc TCS_PT_UNC
                        relative uncertainty of PT, % (default 10)
  --blank BLANK         reagent blank spectrum (prepared and counted like the
                        samples); its activities are subtracted line by line
  --mass MASS           sample quantity
  --unit UNIT           kg, g, L, mL (default kg)
  --mass-unc MASS_UNC   quantity uncertainty %
  --ref-date REF_DATE   reference (sampling) date, default acquisition start
  --nuclides NUCLIDES   comma separated, default Cs-137,K-40,Ra-226,Th-232,Cs-
                        134,Co-60,I-131,Be-7,Am-241
  --method {key,mean}   key line or weighted mean
  --mda {currie,iso11929}
  --attenuation ATTENUATION
                        self-attenuation correction factor
  -o, --out OUT         write results .csv or .xlsx
```

C. Glossary (English – Arabic)

English	Arabic	Meaning
Live time / real time / dead time	الزمن الحي / الحقيقي / الميت	Counting time corrected for dead time / clock time / their difference in %.
Channel, centroid	القناة، مركز القمة	MCA bin; fitted peak position in channels.

English	Arabic	Meaning
Energy calibration	معايرة الطاقة	Polynomial E(channel) in keV.
FWHM	قدرة الفصل (العرض عند نصف الارتفاع)	Full width at half maximum of a peak.
Net / gross area	المساحة الصافية / الإجمالية	Peak counts above / including the background.
Significance	الدلالة الإحصائية	Net area divided by its 1σ uncertainty.
Region of interest (ROI)	منطقة الاهتمام	Marked channel range with gross/net counts.
Overlay	طيف متراكب	Second spectrum drawn on the same plot.
Nuclide identification	التعرّف على النويدات	Matching peaks to library lines.
Confidence	الثقة	Heuristic identification score 0–100 %.
Full-energy-peak efficiency	كفاءة قمة الطاقة الكاملة	Counts in the peak per emitted photon.
Emission probability (yield)	احتمال الانبعاث (الشدة)	Photons per 100 decays.
Specific activity	النشاط النوعي	Bq per kg or per litre.
Expanded uncertainty (k = 2)	الارتياب الموسَّع	2 × combined standard uncertainty (≈ 95 %).
Decision threshold / detection limit	عتبة القرار / حد الكشف	L_C or y* / L_D or y#.
MDA	أدنى نشاط قابل للكشف	Minimum detectable activity.
True-coincidence summing	جمع التوافق الحقيقي	Loss/gain of counts by cascade gamma rays.
Self-attenuation	الامتصاص الذاتي	Absorption of gamma rays inside the sample.
Decay correction	تصحيح الاضمحلال	Activity referred to the sampling date.
Check source, control chart	مصدر الفحص، مخطط التحكم	QC measurement and its chart.
Baseline	خط الأساس	Points that define mean and σ of a control chart.
Gain drift	انجراف الكسب	Slow change of the energy scale.

D. Credits, licences and version

Program	SpectraMaster 1.0.0 — designed & developed by **Edrees F. Hery**.
Qt / PySide6	Graphical interface; © The Qt Company, LGPL v3.
SpecUtils	Reading/writing of many formats (N42, PCF, SPE, CNF, SPC …); © Sandia National Laboratories, LGPL 2.1.
SandiaDecay	Nuclear decay data used to build and cross-check the library; © Sandia National Laboratories, LGPL 2.1.
IAEA ndlab	ENSDF-based decay data (LiveChart of Nuclides database); © IAEA, MIT licence.
NumPy, SciPy, openpyxl	Numerics, fitting and Excel files (BSD / MIT licences).

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Designed & developed by

Edrees F. Hery